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Conference 2000

Panel Session 809P: Healthcare Deployment of Palm Powered Devices

Kevin Maguire
Moderator

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Panel Participants:

- John DiGiovanni: Senior Computer Analyst, Christianna Care Health System, Wilmington Delaware
- Karen Basen-Engquist: Asst. Professor of Behavioral Science, MD Anderson Cancer Center, Houston Texas
- Ray Duncan, MD: Director of Technology, Cedars Sinai Medical Center, Los Angeles California



Healthcare Deployment of Palm OS Devices

John DiGiovanni
Senior Computer Analyst

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Presentation Agenda

- Who are we?
- Our infrastructure
- Palm Standards
- Installation and manageability
  - (Palm HotSync Server)
- Palm Solutions
- Future

Christiana Care Health System

- 1,100 licensed hospital beds in two hospitals
- 8,000 Employees
- A major regional heart center
- Level One Trauma Center
- Eleven family or internal medicine satellite/physician offices
- Home health care provider though VNA
- 99-bed Long-Term and Transitional Care facility

Our Infrastructure

- TCP/IP Ethernet Network
- 230 Windows NT Servers
- 4,500 Plus Windows NT Workstations
- Automated Software Deployment
- Palm HotSync Server
- Approximately 400 Palm Devices

Palm Standards

- Standardized on Palm IIIxe, Vx and IIIc
- HotSync Server and IR Syncing Devices
- Created process for deploying Palm™ handhelds to our users (to maintain consistency)
- Trained Help Desk and Field Support Staff
- In-house replacement process

Software Installation

Windows NT Client

- Automatic installation of software based on NT Group Membership
  - Palm HotSync Manager
  - Chapura NomadSync Client Software
  - HotSync® Server Client and Desktop Proxy Agent

Software/User Management

Palm Applications

- Automatic installation of software based on HotSync® Server Group Membership
- Palm OS Upgrades

User Management

- SQL Server Based User Profiles
- User Backups to server
- NT Remote Administration/Support

Palm Solutions

Shareware Applications

- ePocrates
- MedMath
- PalmPrint
- MS Word/MS Excel Readers
- QuickNotes
- AvantGo

Palm Solutions

Custom Forms

- OB Resident Procedure Tracking
- Surgery Resident Procedure Tracking
- Pharmacy Inventory Tracking

Palm Future

- Web Enabled Applications
- Increase usage of HotSync Server with conduits
  - Additional Outlook/Exchange functionality
  - HTML/XML/ASP Support
  - Database Synchronization
  - Server-based security management
- Future Government HIPAA Requirements
  - (Health Information Portability & Accountability Act of 1996)

Thank You!

Contact Information

- www.ChristianaCare.org
- John DiGiovanni
  - jdigiovanni@christianacare.org

Handheld Computers in Health Research

THE UNIVERSITY OF TEXAS
MD ANDERSON
CANCER CENTER

Karen Basen-Engquist, Ph.D.
Assistant Professor of
Behavioral Science

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Presentation Overview

- Context
- How behavioral health research is usually done
- How handheld computers can change that
- Illustration from an ongoing project
- Other applications in behavioral health research at M. D. Anderson

Context: M. D. Anderson Cancer Center

- Comprehensive Cancer Center
- Mission includes patient care, research, education, and prevention
- 1,575 outpatient visits, 347 inpatients daily
- Department of Behavioral Science—research department
- Areas of research include tobacco use and cancer patient quality of life



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M. D. Anderson Cancer Center



Behavioral Health Assessment

- Usually we use retrospective reporting over a specific time period
- Example:

Please indicate how true each statement has been for you during the past 7 days.

	Not at all	A little bit	Somewhat	Very much
1. I have pain.	_____	_____	_____	_____
2. I have lack of energy.	_____	_____	_____	_____

Problems with Traditional Approaches

- Recall
  - Some things may be forgotten
  - Others may be recalled in a biased way
- Aggregating over the time period
- Inaccuracies can be introduced by the testing situation, current emotions
- Long time periods don't provide information on short-term changes, e.g., daily patterns
- One assessment point provides no evidence of the direction of causation

Solution: Real Time, Real World Assessment

- Termed Ecological Momentary Assessment (EMA)
- May be more accurate than information collected “after the fact”
- Data collected in a person’s environment more representative of his/her actual behaviors and feelings
- Can identify how events vary over time and identify events that precede, coincide, or follow the phenomena of interest

Using Handheld Computers for EMA

- Can be carried easily and used in a person's natural environment
- Combine prompting of data entry, presenting questions, and recording data in one device
- Compliance can be monitored by recording date and time the data are entered
- Questions can be tailored to the participant
- Easy to use

Ecological Momentary Assessment of Fatigue and Other Symptoms of Ovarian Cancer Patients

Behavioral Science

- Karen Basen-Engquist, Ph.D.
- Carl de Moor, Ph.D.
- Cindy Carmack, Ph.D.
- Lorenzo Cohen, Ph.D.
- Mary A. Fitzgerald, M.A.

Analysts International

- Kent Barnes
- Udora Phillips

Gynecologic Oncology

- Diane Bodurka-Bevers, M.D.
- Judith Wolf, M.D.
- Charles Levenback, M.D.

Medical Oncology

- Andrzej Kudelka, M.D.

Internal Medicine

- Carmen Escalante, M.D.

Project Objectives

- To pilot test use of handheld computers to record fatigue and other symptoms of women with ovarian cancer
- To evaluate whether fatigue can be predicted by physical activity, mood, pain, and nausea
- To identify more efficient ways to sample time periods and collect data

Fatigue

- Common symptom/treatment side effect
- Reported by patients receiving all types of treatment, surgery, chemotherapy, radiation, immune therapy
- Cause uncertain
  - Physical factors—anemia, thyroid disease
  - Emotional factors—depression
  - Behavioral factors—physical activity (too much? too little?), sleep disruption, nutrition

Ovarian Cancer

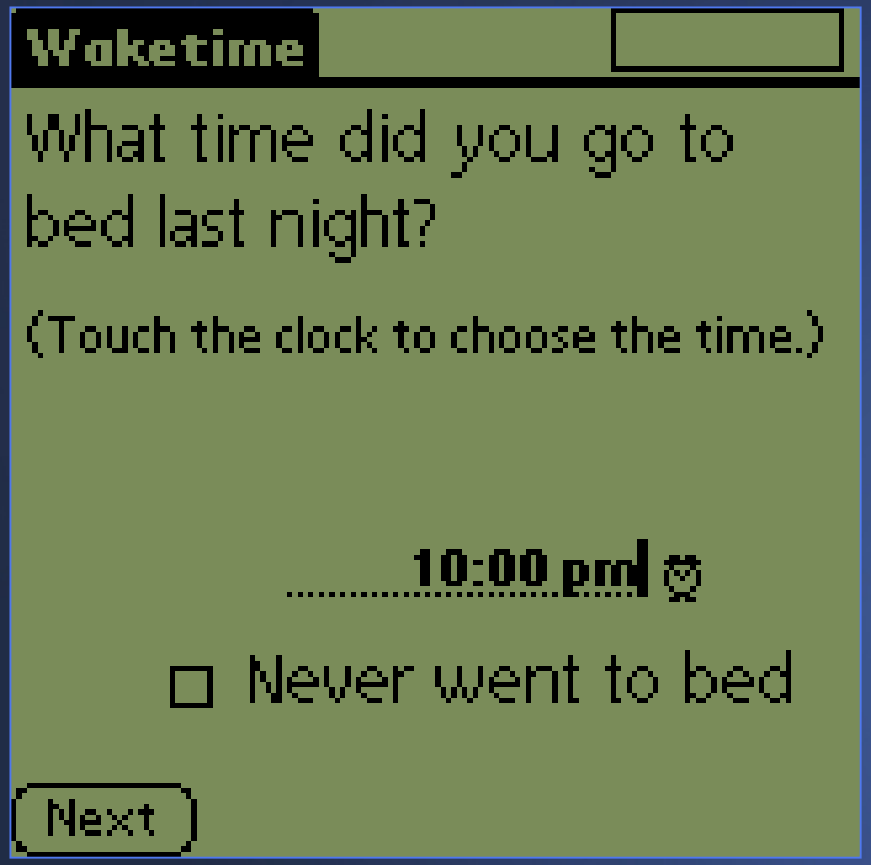
- Most cases are diagnosed when the disease has spread beyond the ovary
- Five-year survival rate is less than 30%
- Initial treatment is surgery, chemotherapy (high dose with bone marrow/stem cell transplant on experimental protocols)
- Recurrence is frequent in patients with advanced disease
- Fatigue is frequently experienced during and after treatment

Study Methods

- Assessments completed at several times during the day:
  - After waking
  - Random assessments during the day
  - Before going to bed
- Asked about sleep, activity, mood, pain and nausea, and fatigue



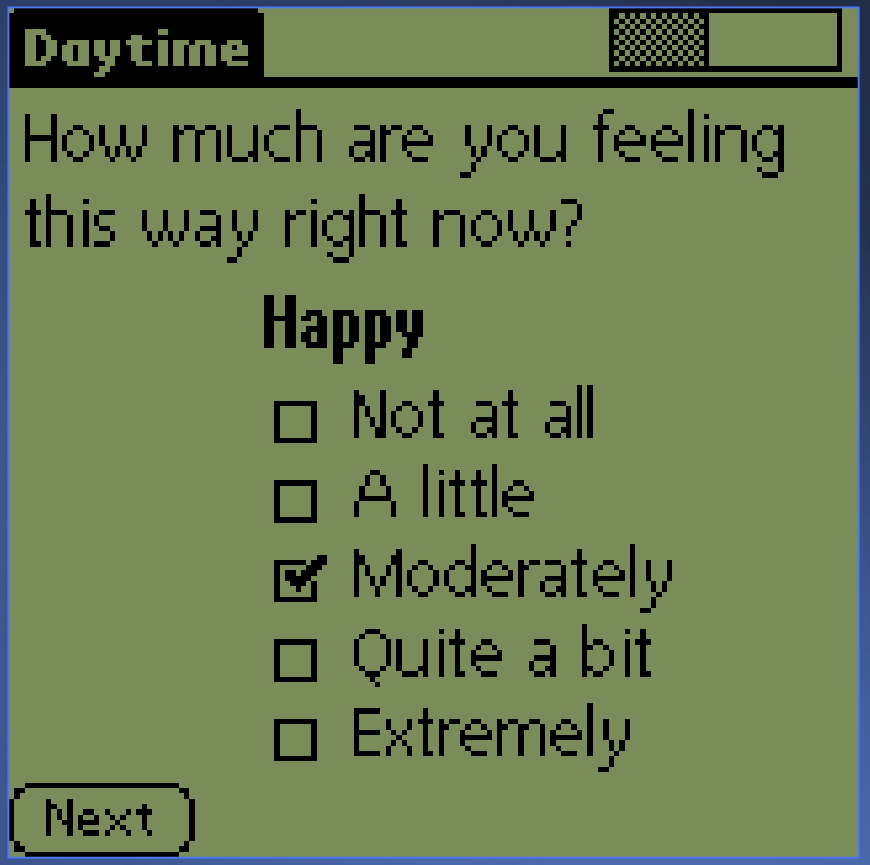
Waketime Assessment



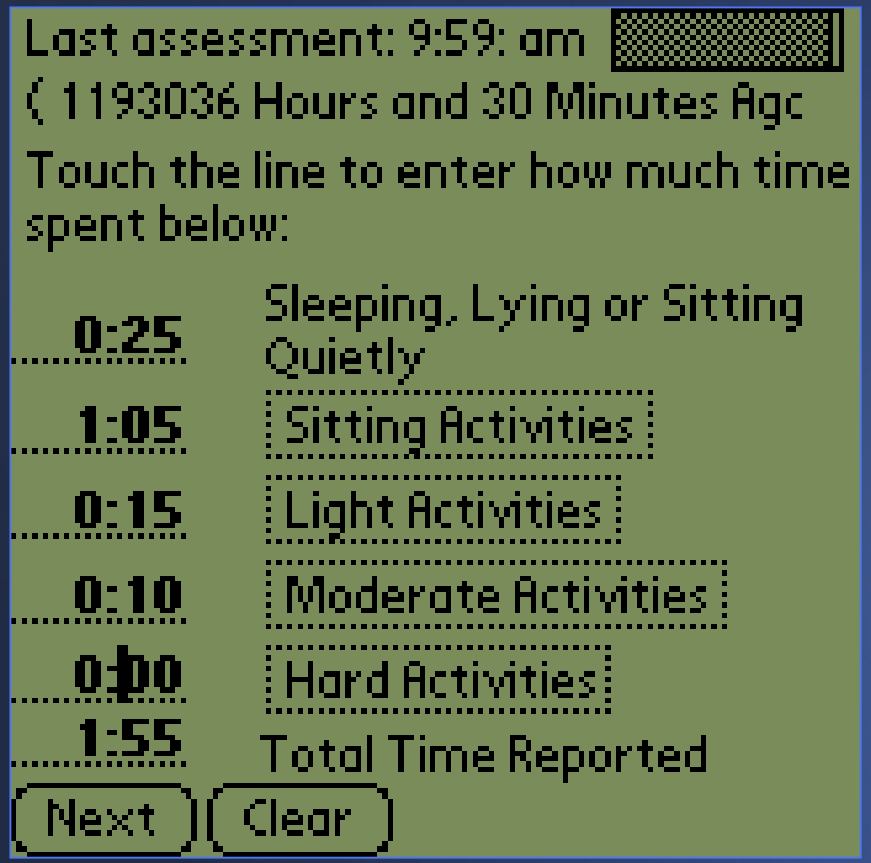


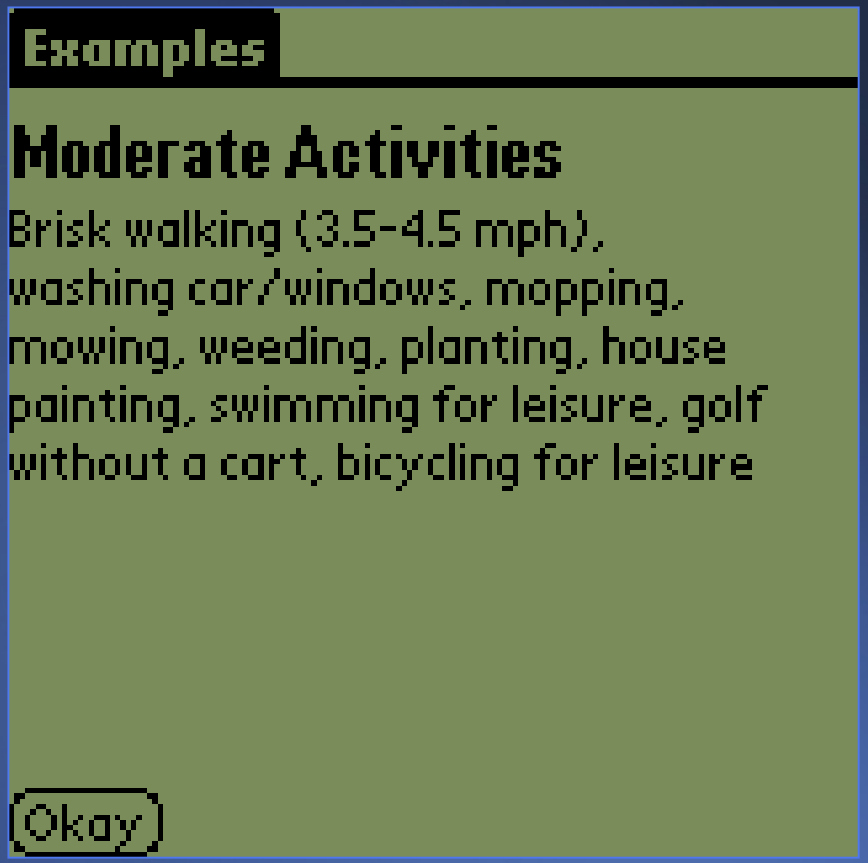
Daytime Assessment



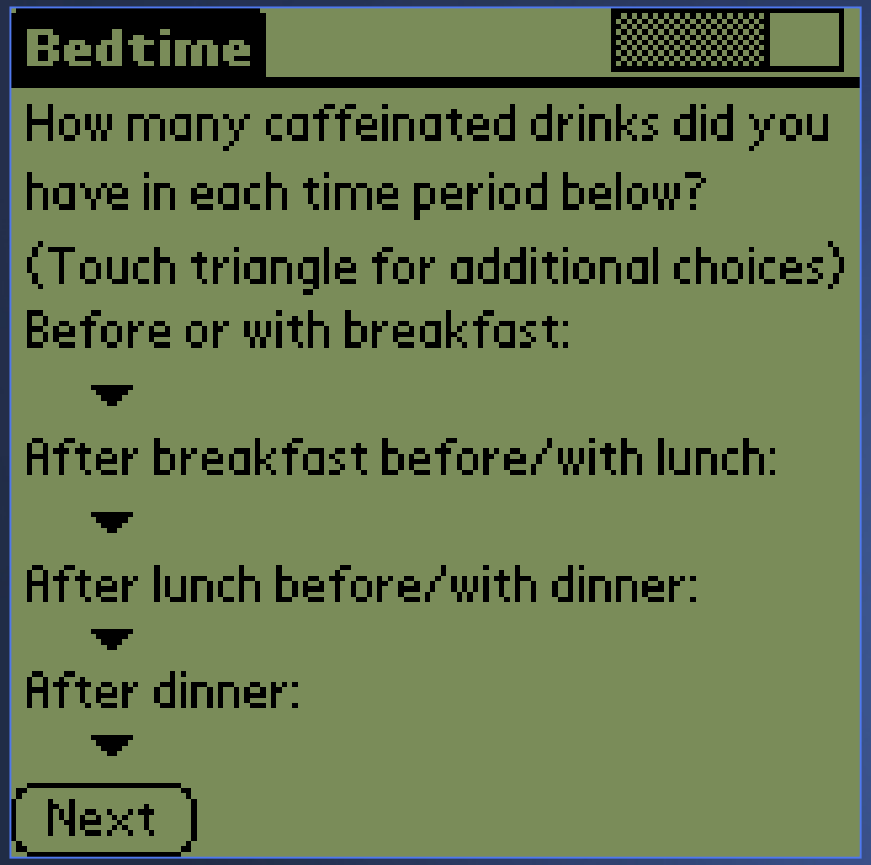


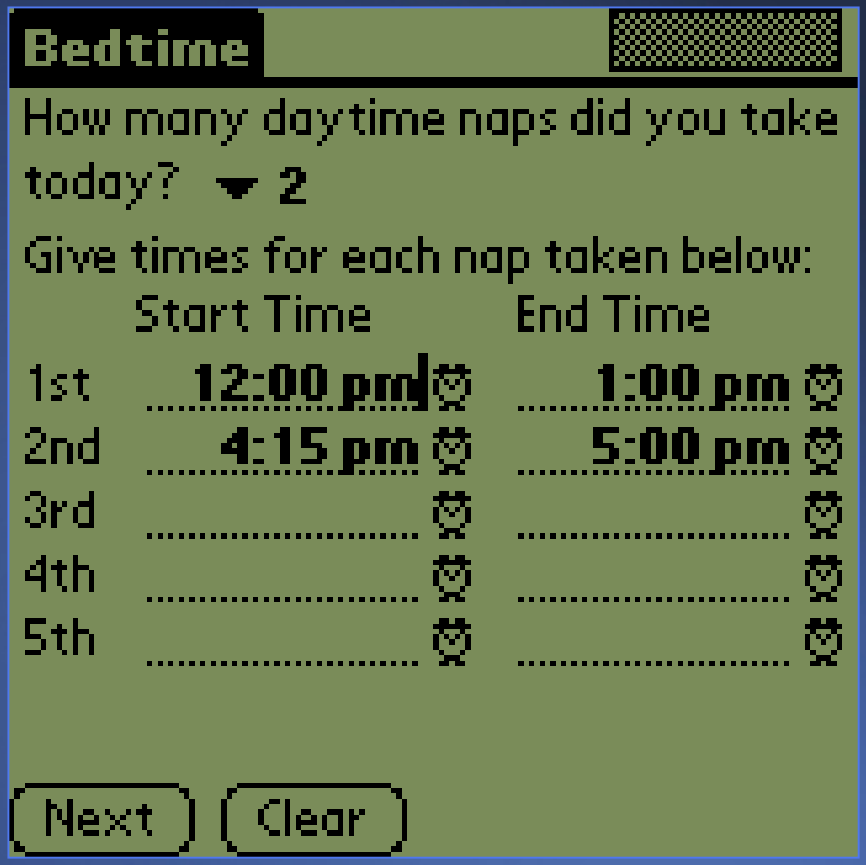
Daytime Assessment



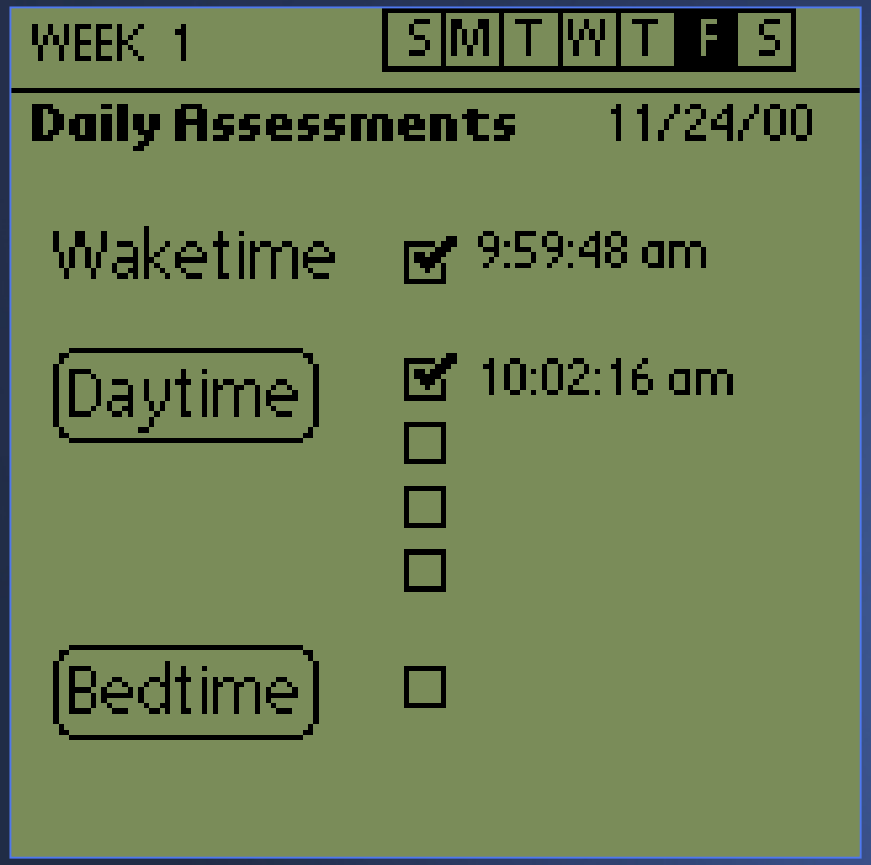


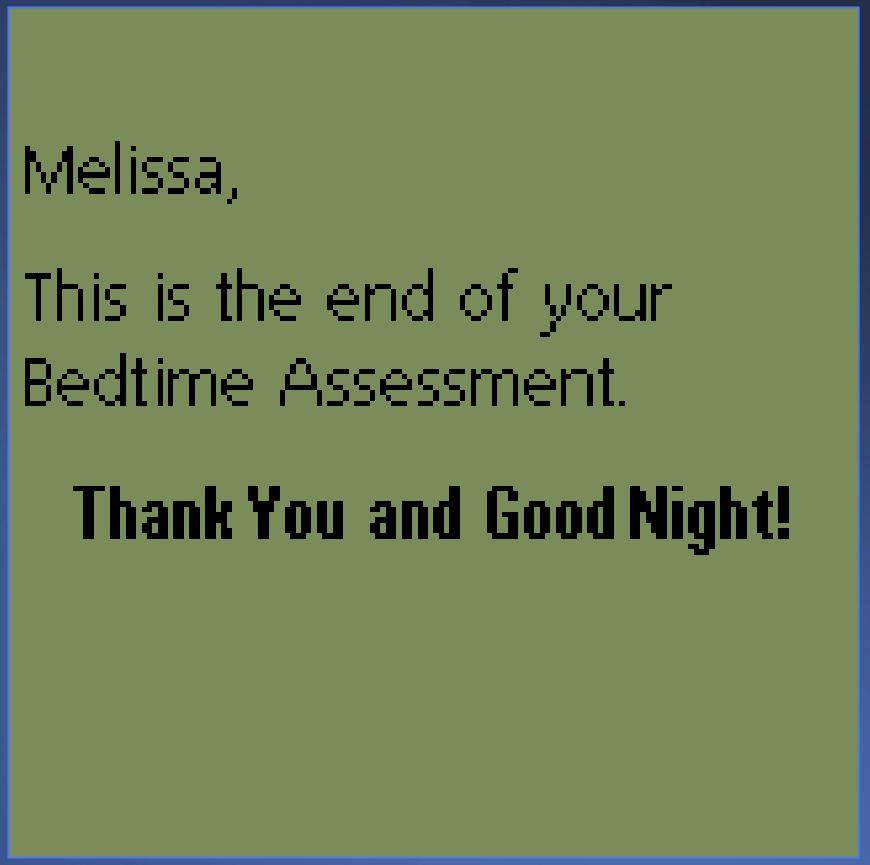
Bedtime Assessment



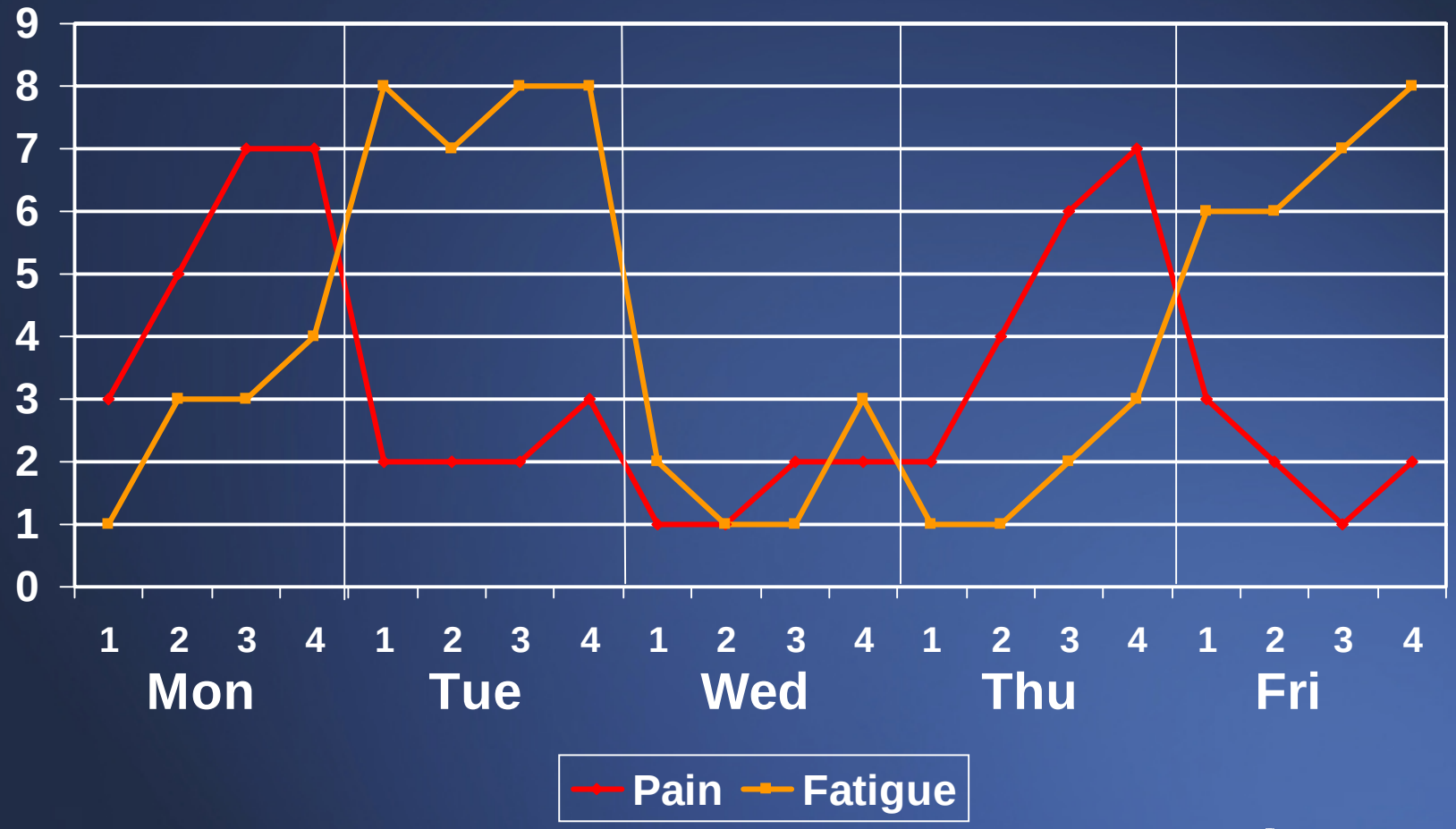


Introductory and Ending Screens





Mock Results—Relationship Between Pain and Fatigue, Patient A



Advantages of EMA for Studying Fatigue

- More precise and accurate measurement
- Uncover daily patterns in fatigue and other symptoms, and identify patterns over the chemotherapy cycle (~3 weeks)
- Able to assess causal relationships
- Information can be used to develop programs to improve energy levels



Other Departmental Projects Using Handheld Computers

Current Projects

- Stress management program for women with breast cancer
- Studies of smoking cessation programs

Future Projects?

- Fatigue management using physical activity
- Symptom management

Secure Wireless Access to Clinical Data for Physicians

Ray Duncan, MD
Director, Technology and Architecture
Enterprise Information Systems
Cedars-Sinai Health System

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Background



- Cedars-Sinai Medical Center is an 820 bed private nonprofit tertiary-care hospital in Los Angeles, CA
- Affiliated with UCLA Medical School, also has free-standing residency and fellowship training programs
- Business unit of Cedars-Sinai Health System
- 10,000 employees, 200+ staff physicians, 250 interns, residents and fellows, 2000 private physicians
- Not just “the hospital to the stars”

Why Palm VII?

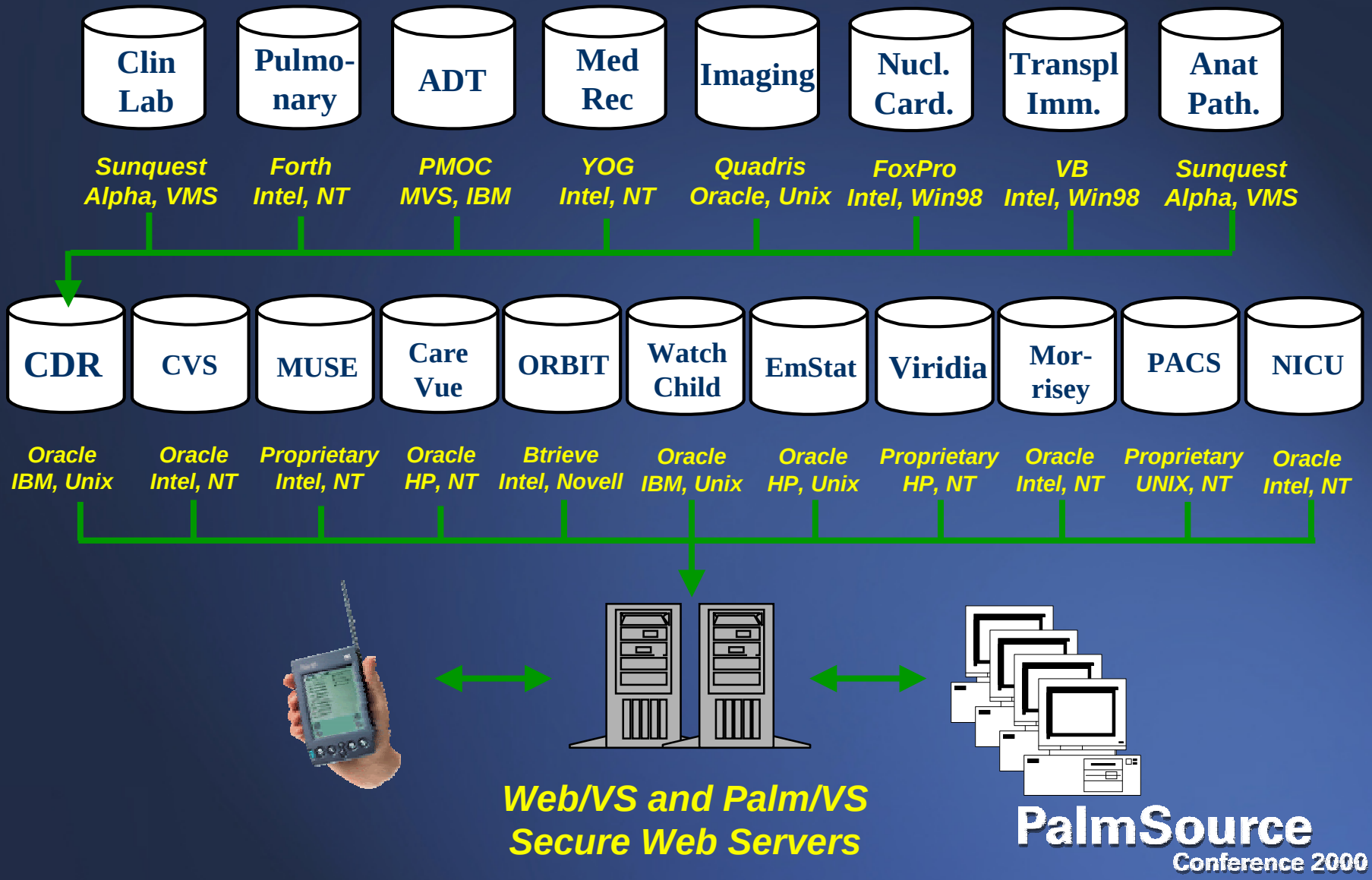
- Out of the box single-vendor solution for PDA with wireless networking
- Excellent user interface and battery life, lots of applications available
- PC and Mac compatible
- Easy conversion of existing web apps
- Previous PDA/wireless projects:
  - HP 200LX plus wireless modem for ICUs
  - Alphanumeric pagers for critical lab values
  - HP Jornada 680 in Pediatric clinic and inpatient ward



Palm/VS—Wireless Access to Clinical Data for Physicians

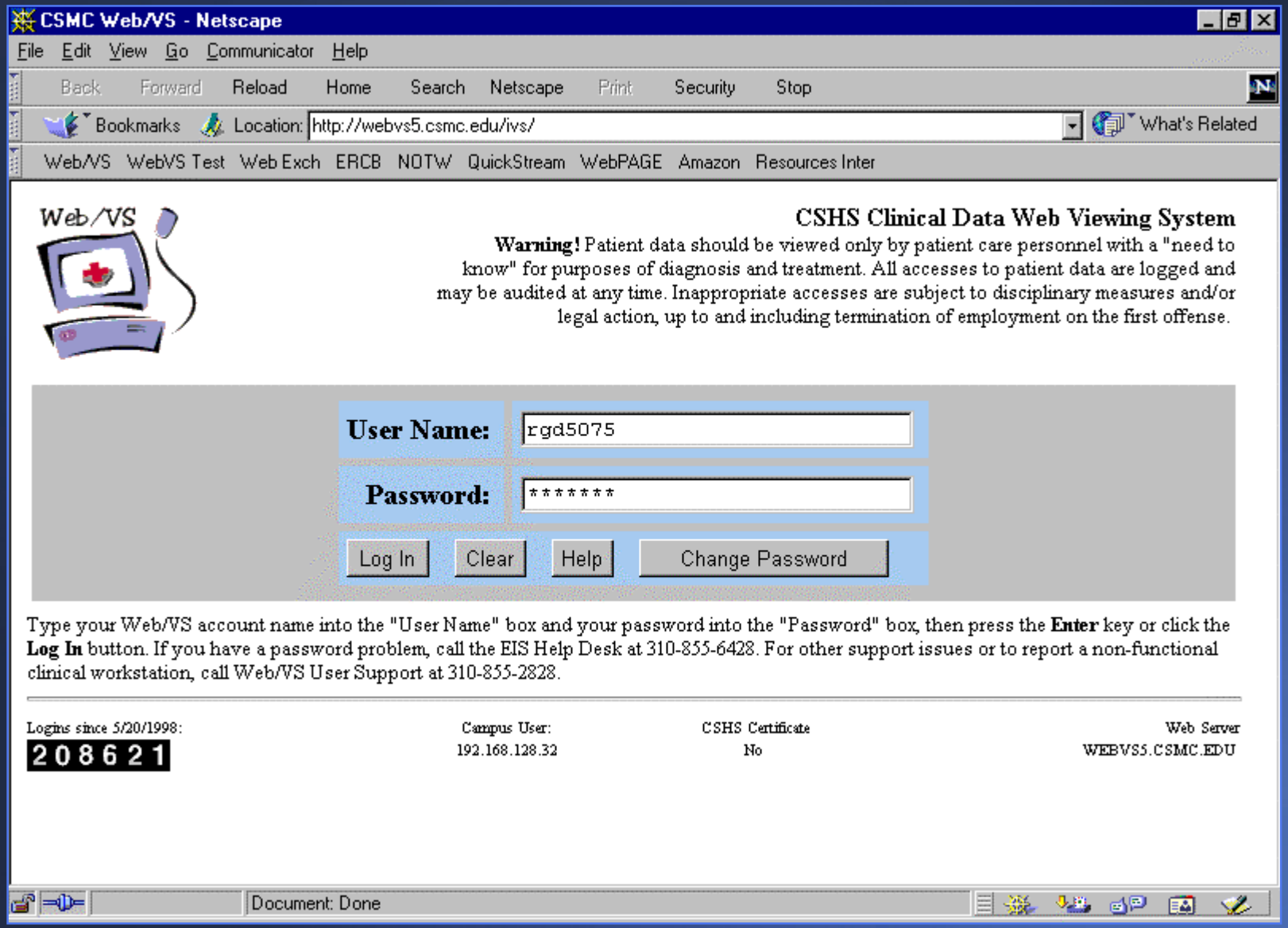
- Palm/VS is derived from a browser-based clinical viewing system (Web/VS) in use housewide since 1998
- Runs on same web servers and accesses same databases as Web/VS
- Subset of Web/VS functionality and data types
- Palm/VS is primarily used for remote access to clinical data

Palm/VS and Web/VS Architecture

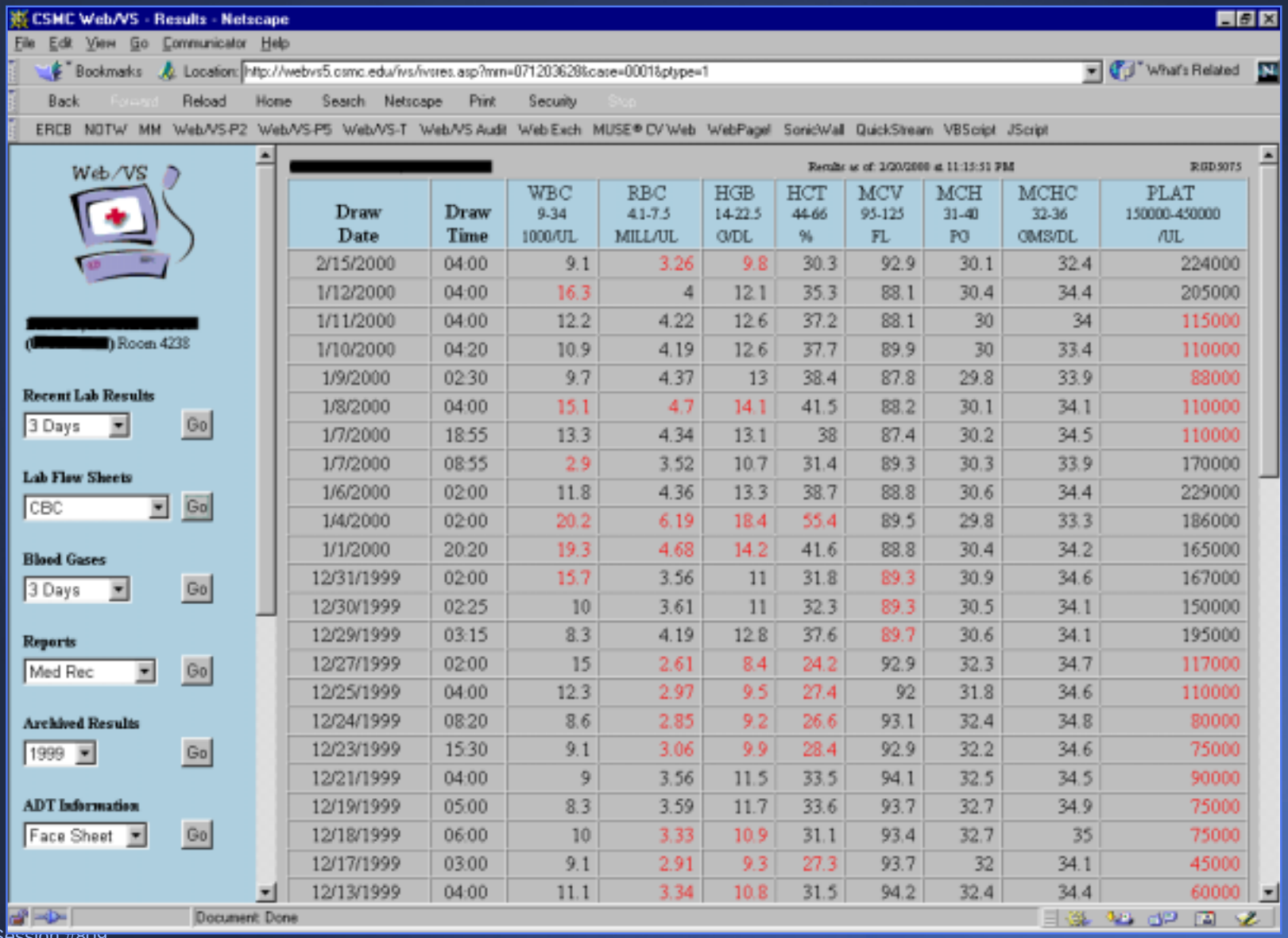


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Web/VS—Log In

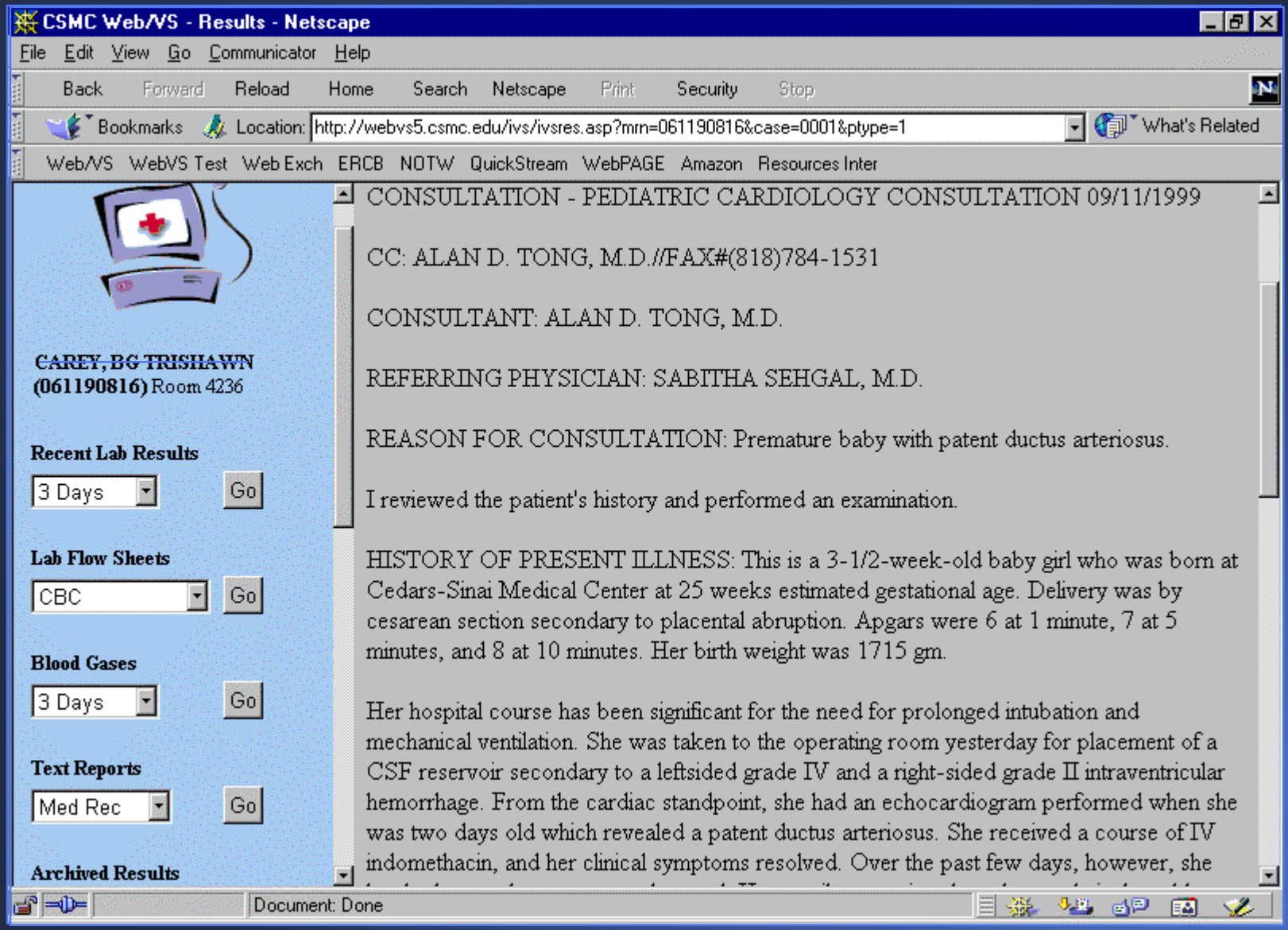

CSMC Web/VS - Netscape
File Edit View Go Communicator Help
Back Forward Reload Home Search Netscape Print Security Stop
Bookmarks Location: http://webvs5.csmc.edu/ivs/
What's Related
Web/VS WebVS Test Web Exch ERCB NOTW QuickStream WebPAGE Amazon Resources Inter
Web/VS
CSHS Clinical Data Web Viewing System
Warning! Patient data should be viewed only by patient care personnel with a "need to know" for purposes of diagnosis and treatment. All accesses to patient data are logged and may be audited at any time. Inappropriate accesses are subject to disciplinary measures and/or legal action, up to and including termination of employment on the first offense.
User Name: rgd5075
Password: *******
Log In
Clear
Help
Change Password
Type your Web/VS account name into the "User Name" box and your password into the "Password" box, then press the Enter key or click the Log In button. If you have a password problem, call the EIS Help Desk at 310-855-6428. For other support issues or to report a non-functional clinical workstation, call Web/VS User Support at 310-855-2828.
Logins since 5/20/1998:
208621
Campus User:
192.168.128.32
CSHS Certificate
No
Web Server
WEBVS5.CSMC.EDU
Document: Done

Web/VS—Flowsheet Views

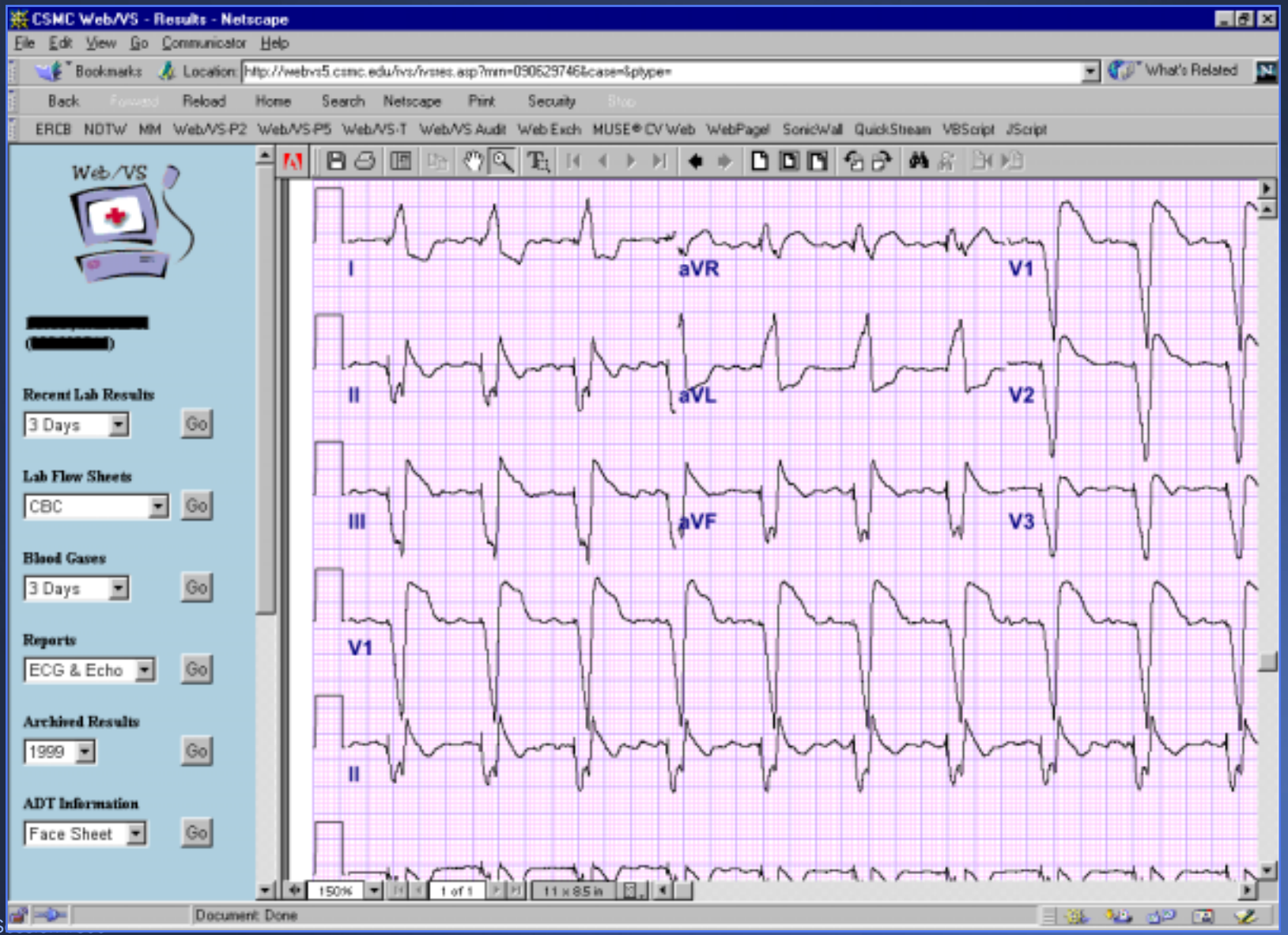


Draw Date	Draw Time	WBC 9-34 1000/UL	RBC 4.1-7.5 MILL/UL	HGB 14-22.5 G/DL	HCT 44-66 %	MCV 95-125 FL	MCH 31-40 PG	MCHC 32-36 GMS/DL	PLAT 150000-450000 /UL
2/15/2000	04:00	9.1	3.26	9.8	30.3	92.9	30.1	32.4	224000
1/12/2000	04:00	16.3	4	12.1	35.3	88.1	30.4	34.4	205000
1/11/2000	04:00	12.2	4.22	12.6	37.2	88.1	30	34	115000
1/10/2000	04:20	10.9	4.19	12.6	37.7	89.9	30	33.4	110000
1/9/2000	02:30	9.7	4.37	13	38.4	87.8	29.8	33.9	88000
1/8/2000	04:00	15.1	4.7	14.1	41.5	88.2	30.1	34.1	110000
1/7/2000	18:55	13.3	4.34	13.1	38	87.4	30.2	34.5	110000
1/7/2000	08:55	2.9	3.52	10.7	31.4	89.3	30.3	33.9	170000
1/6/2000	02:00	11.8	4.36	13.3	38.7	88.8	30.6	34.4	229000
1/4/2000	02:00	20.2	6.19	18.4	55.4	89.5	29.8	33.3	186000
1/1/2000	20:20	19.3	4.68	14.2	41.6	88.8	30.4	34.2	165000
12/31/1999	02:00	15.7	3.56	11	31.8	89.3	30.9	34.6	167000
12/30/1999	02:25	10	3.61	11	32.3	89.3	30.5	34.1	150000
12/29/1999	03:15	8.3	4.19	12.8	37.6	89.7	30.6	34.1	195000
12/27/1999	02:00	15	2.61	8.4	24.2	92.9	32.3	34.7	117000
12/25/1999	04:00	12.3	2.97	9.5	27.4	92	31.8	34.6	110000
12/24/1999	08:20	8.6	2.85	9.2	26.6	93.1	32.4	34.8	80000
12/23/1999	15:30	9.1	3.06	9.9	28.4	92.9	32.2	34.6	75000
12/21/1999	04:00	9	3.56	11.5	33.5	94.1	32.5	34.5	90000
12/19/1999	05:00	8.3	3.59	11.7	33.6	93.7	32.7	34.9	75000
12/18/1999	06:00	10	3.33	10.9	31.1	93.4	32.7	35	75000
12/17/1999	03:00	9.1	2.91	9.3	27.3	93.7	32	34.1	45000
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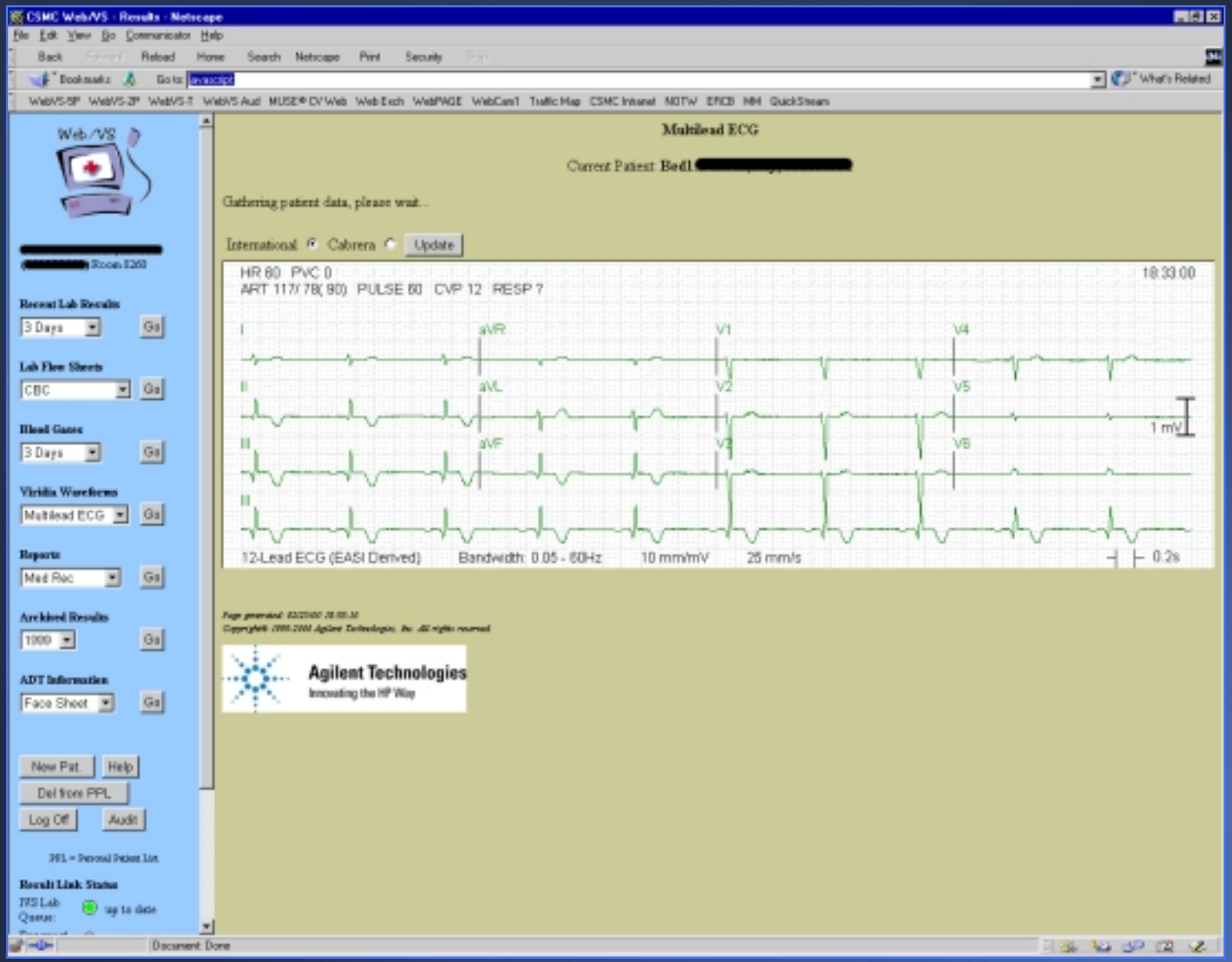
Web/VS—Transcribed Reports



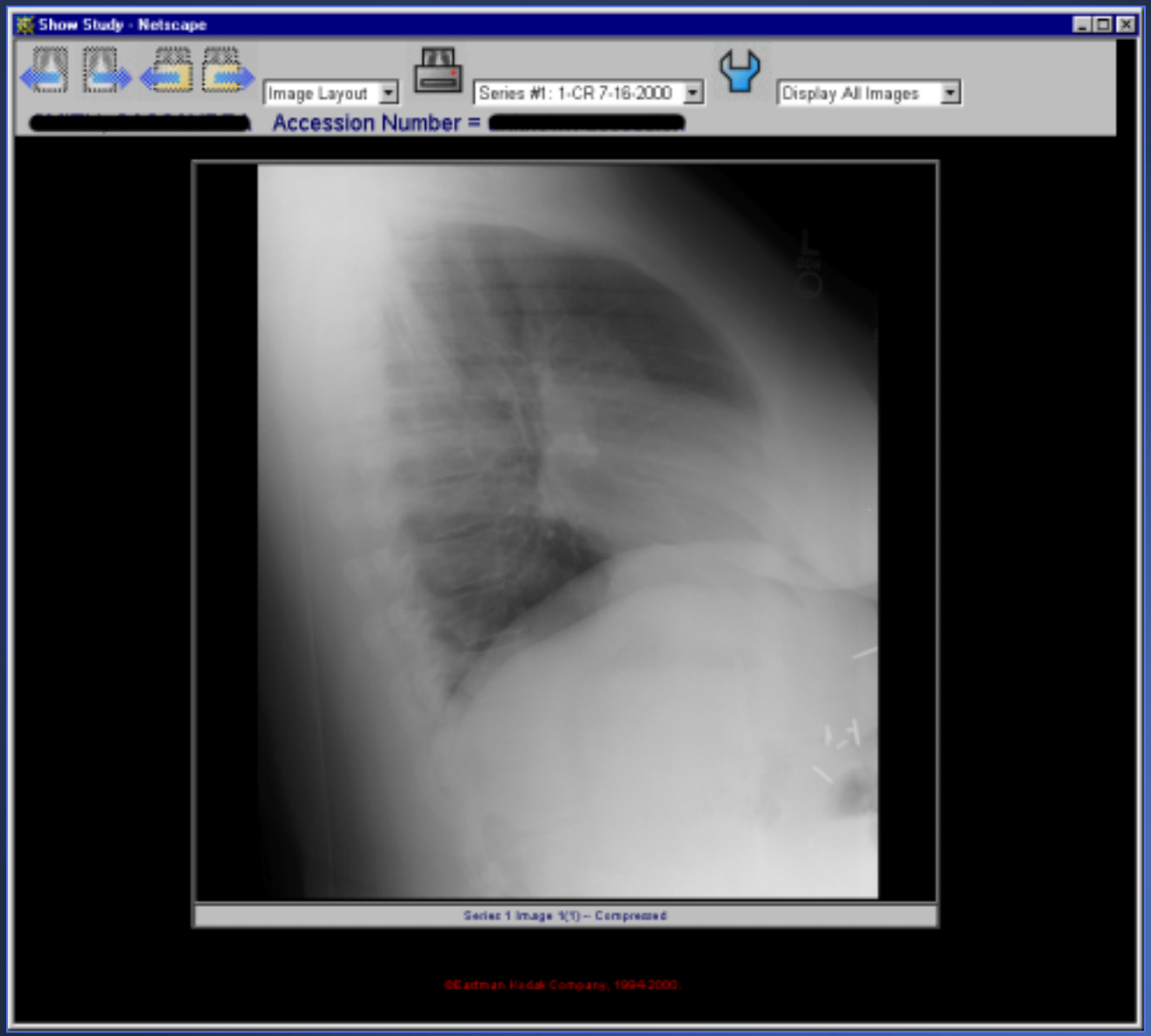
Web/VS—ECG Reports



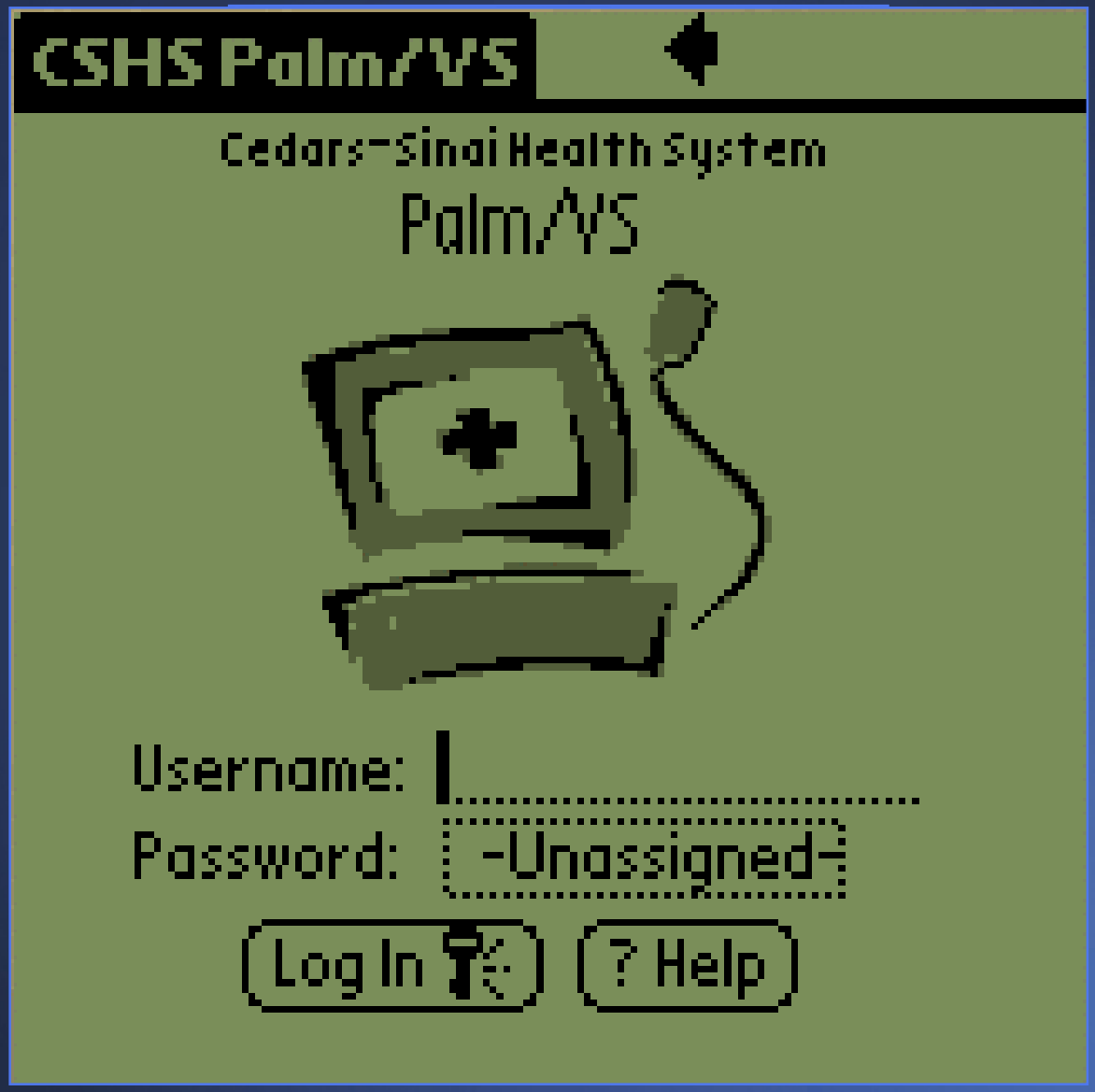
Web/VS—ICU Monitor Waveforms



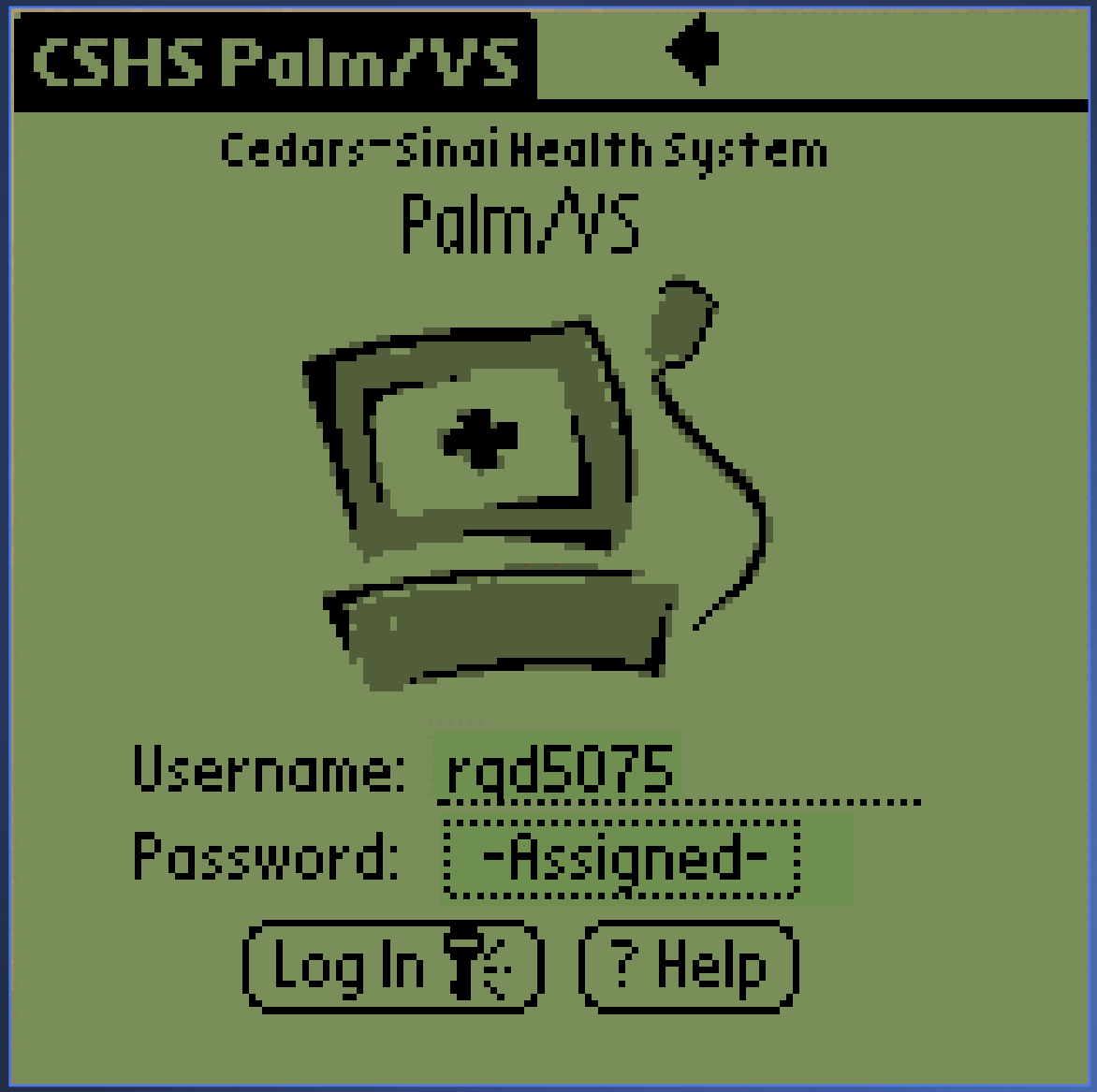
Web/VS—PACS Images



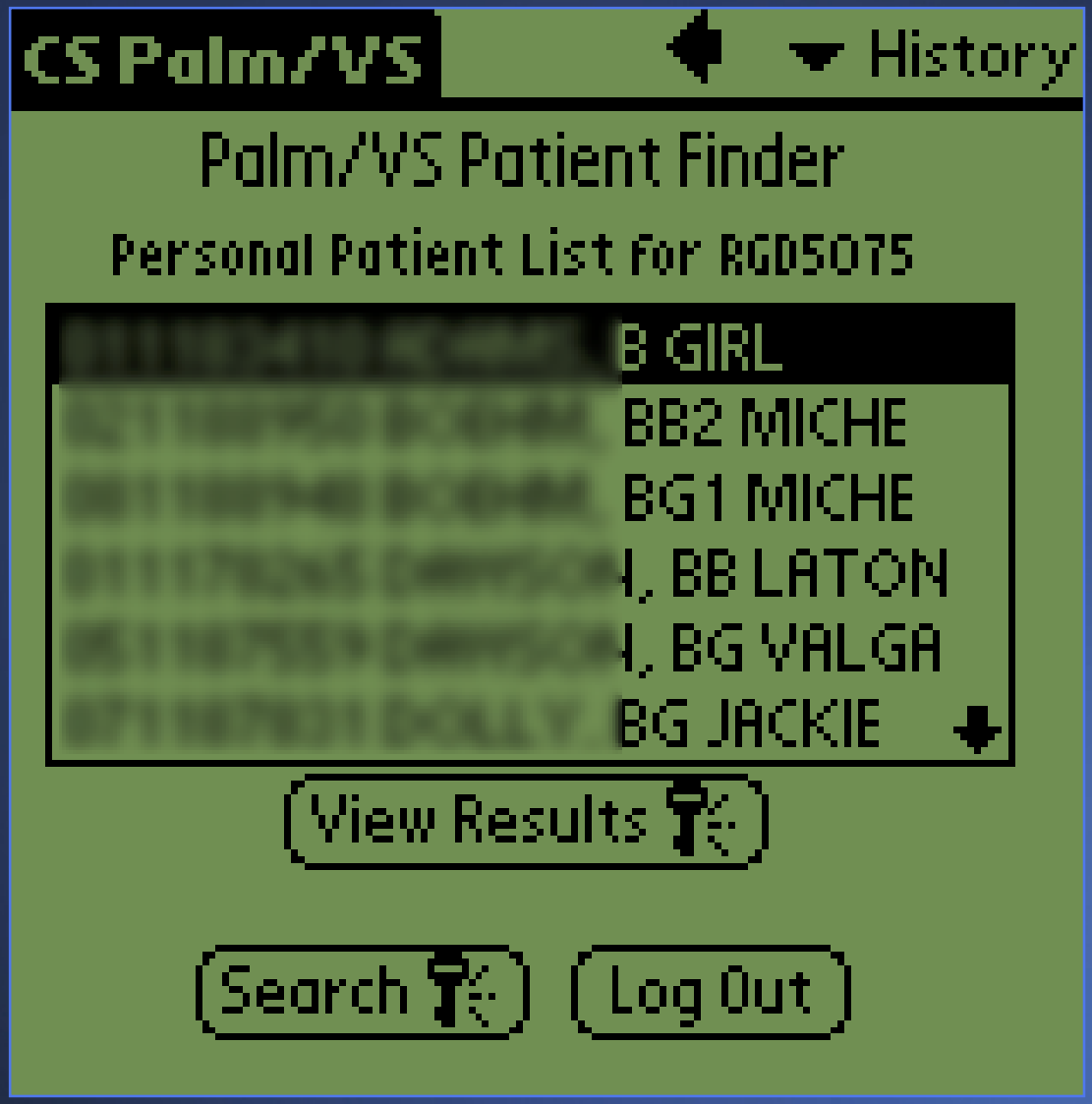
Palm/VS—Clinical Data Viewer



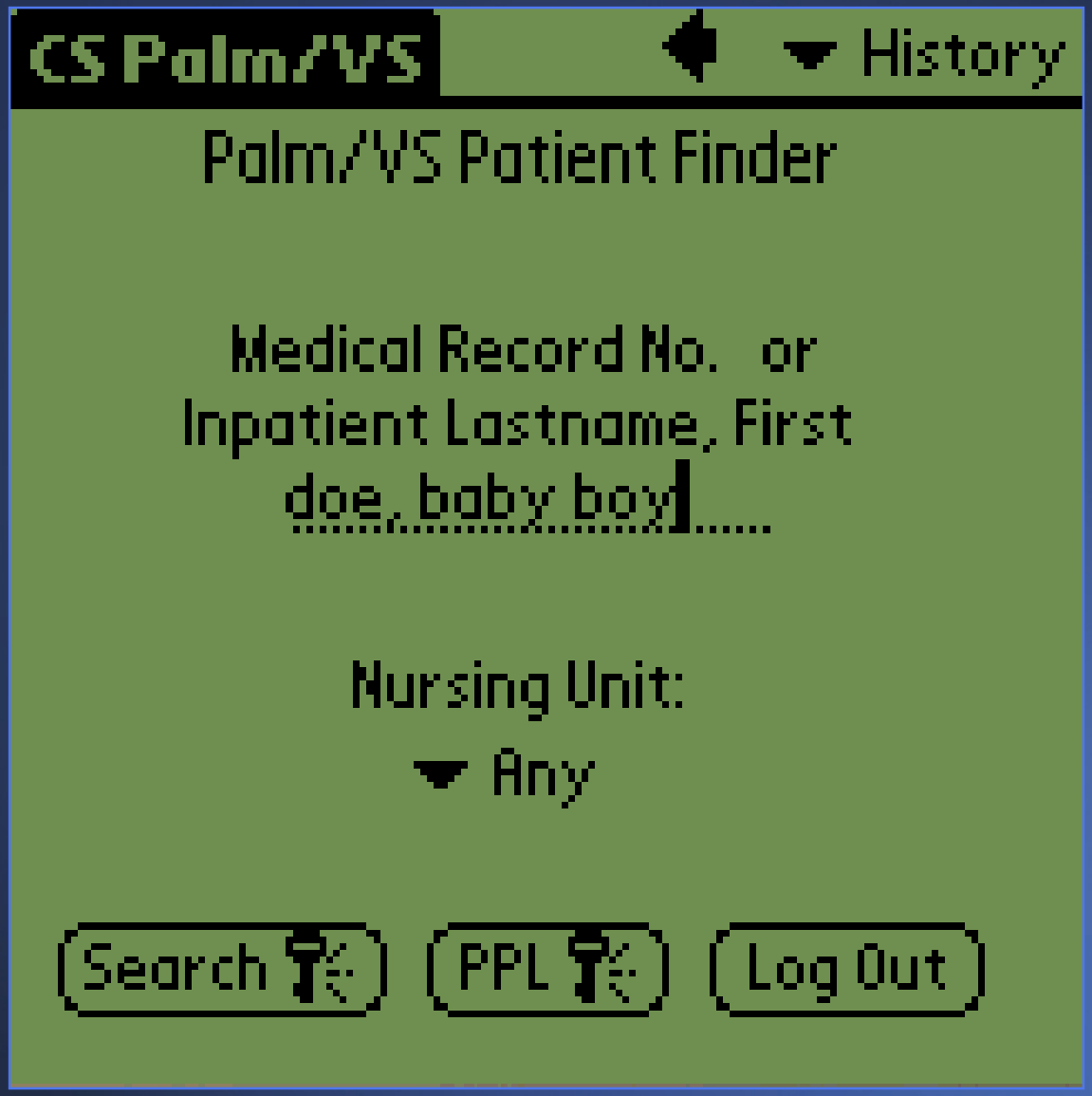
Palm/VS—Log In



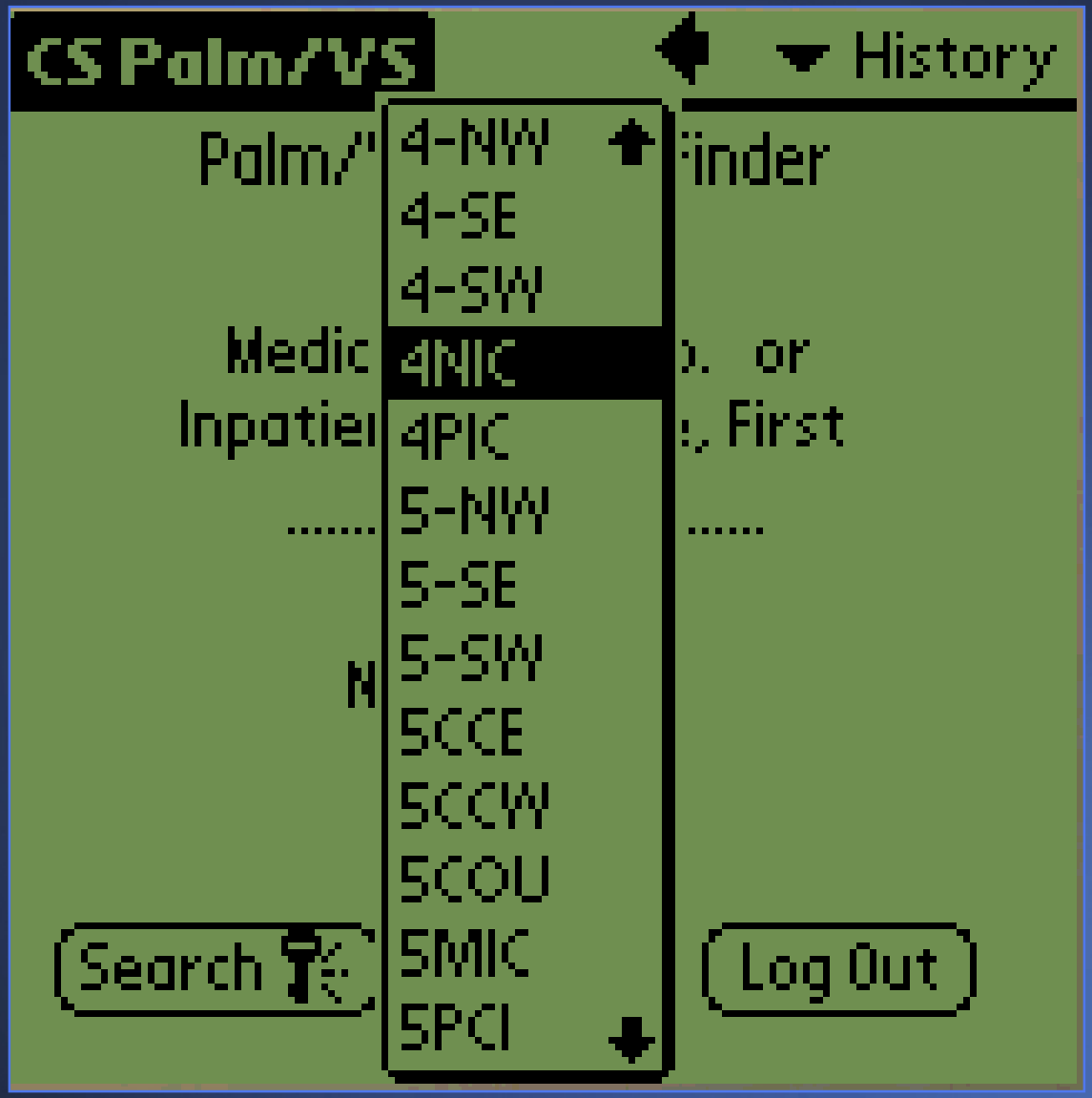
Palm/VS—Personal Patient List



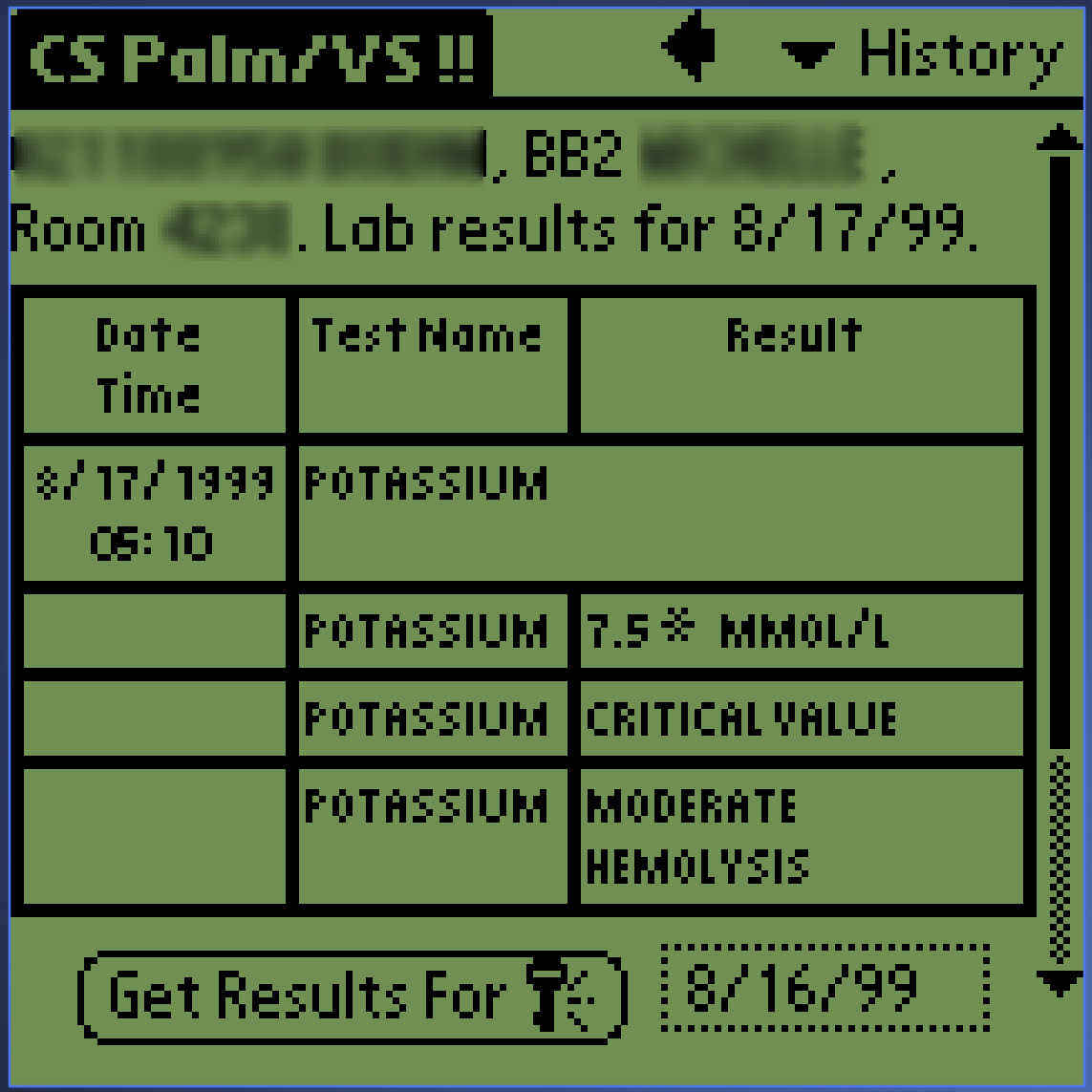
Palm/VS—Search by Name



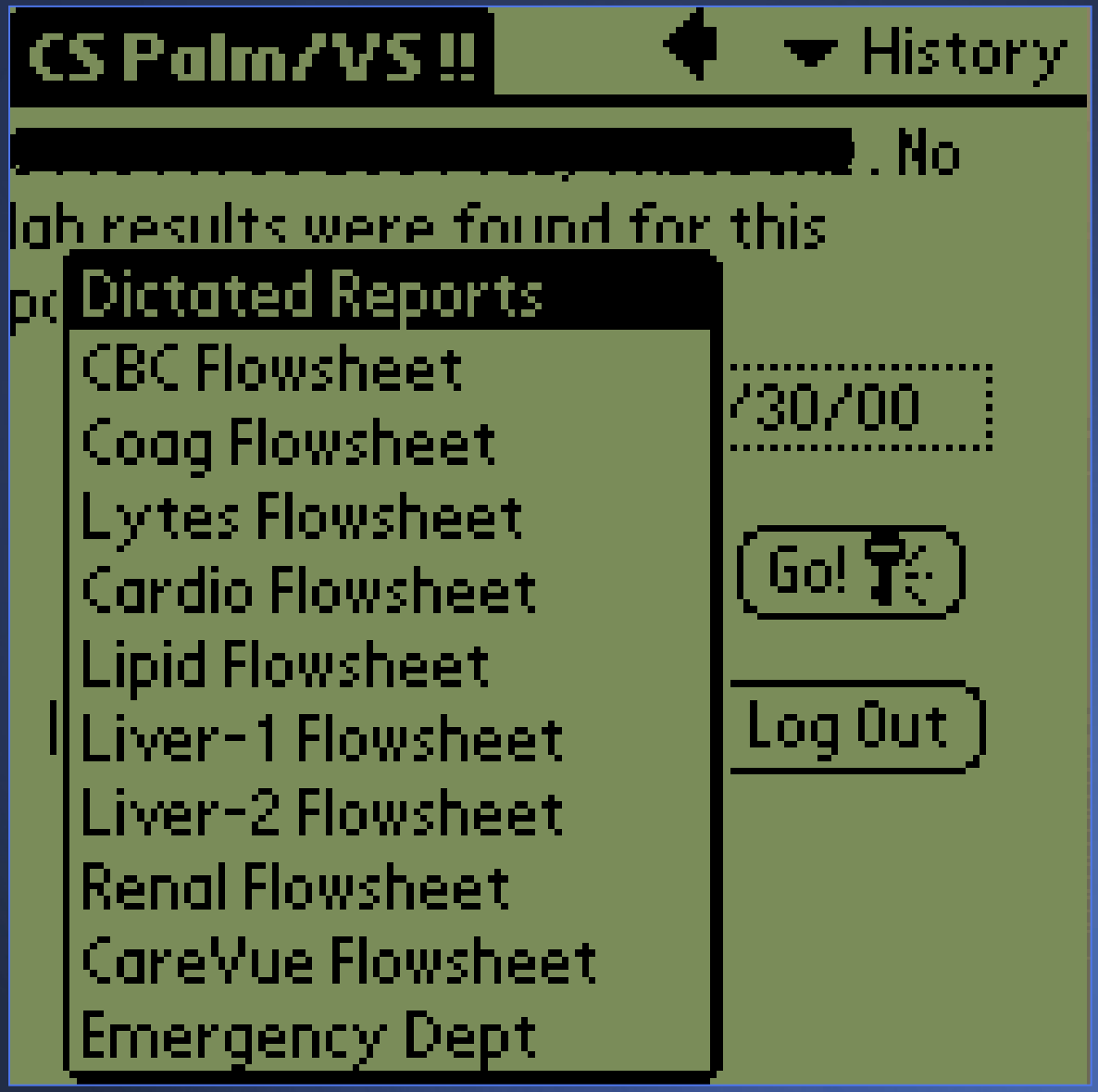
Palm/VS—Search by Unit



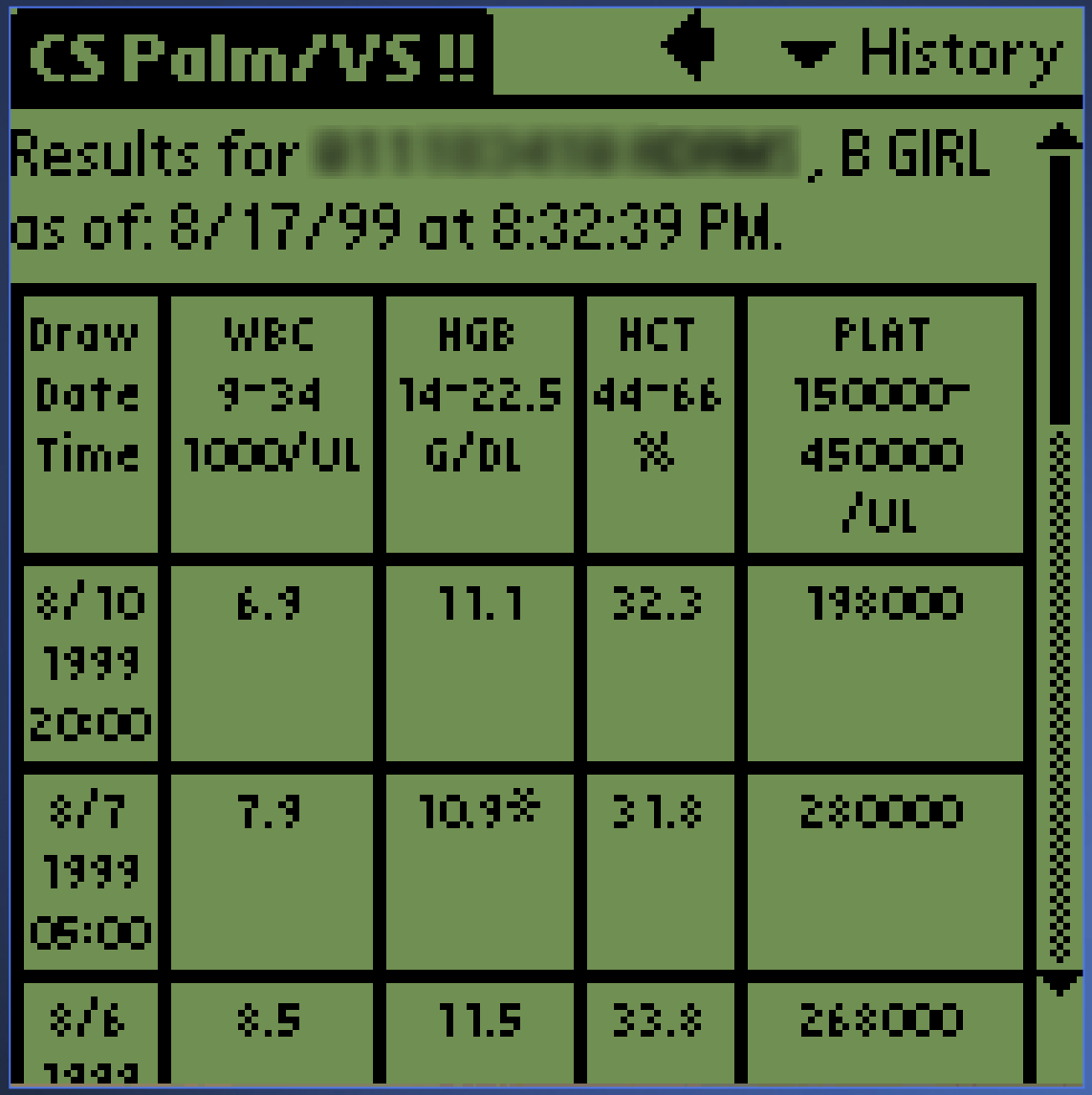
Palm/VS—Recent Results



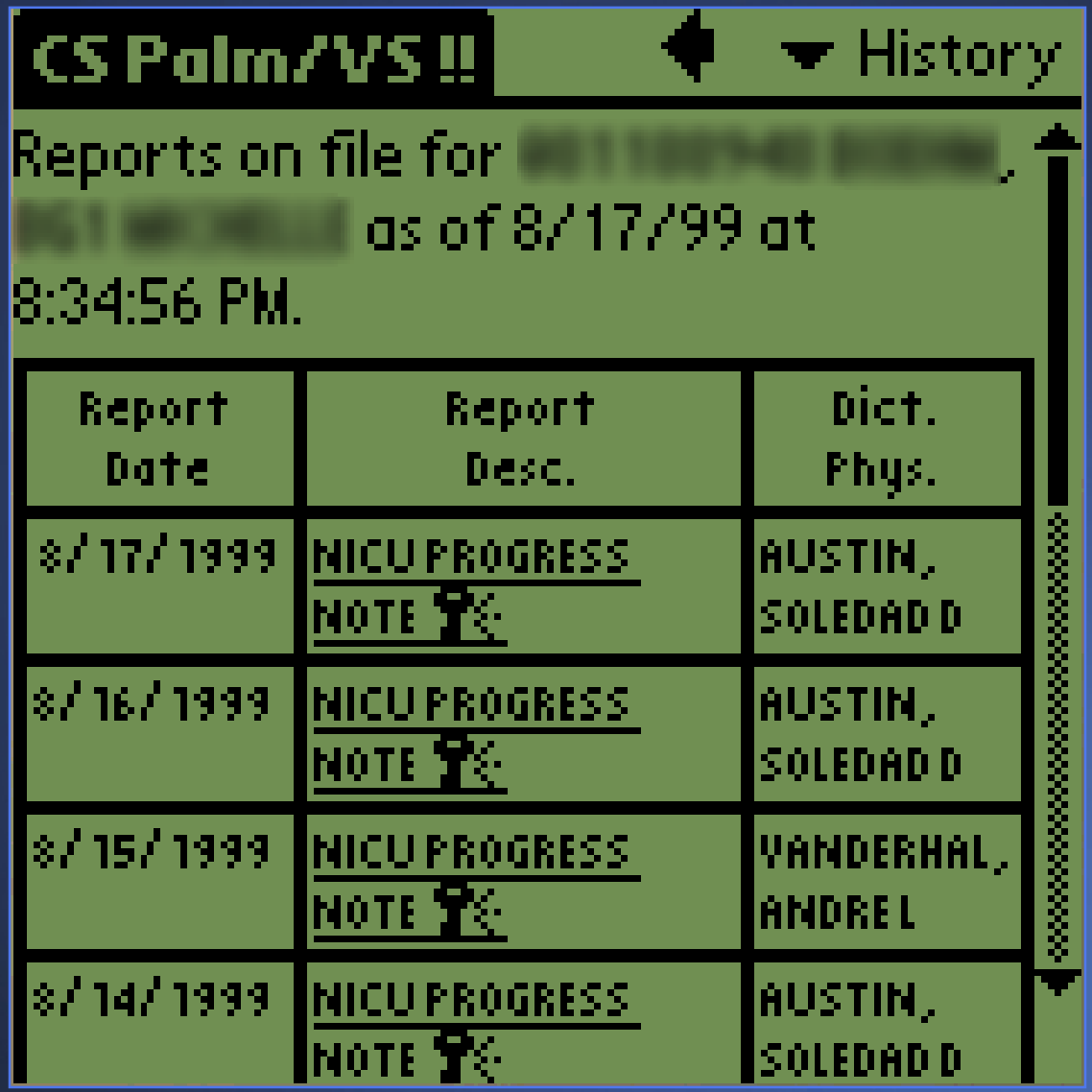
Palm/VS—Reports Menu



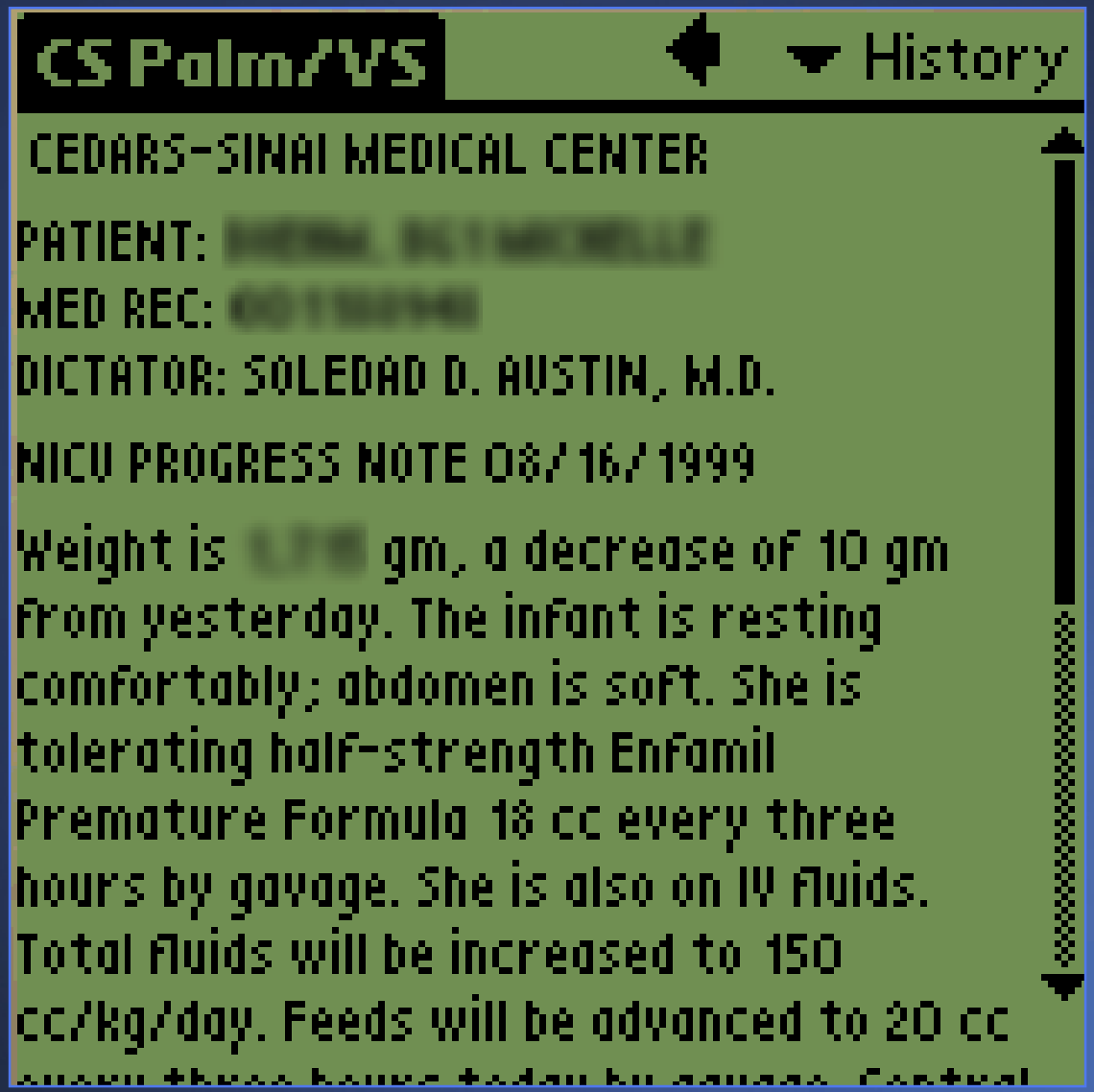
Palm/VS—Sample Flowsheet



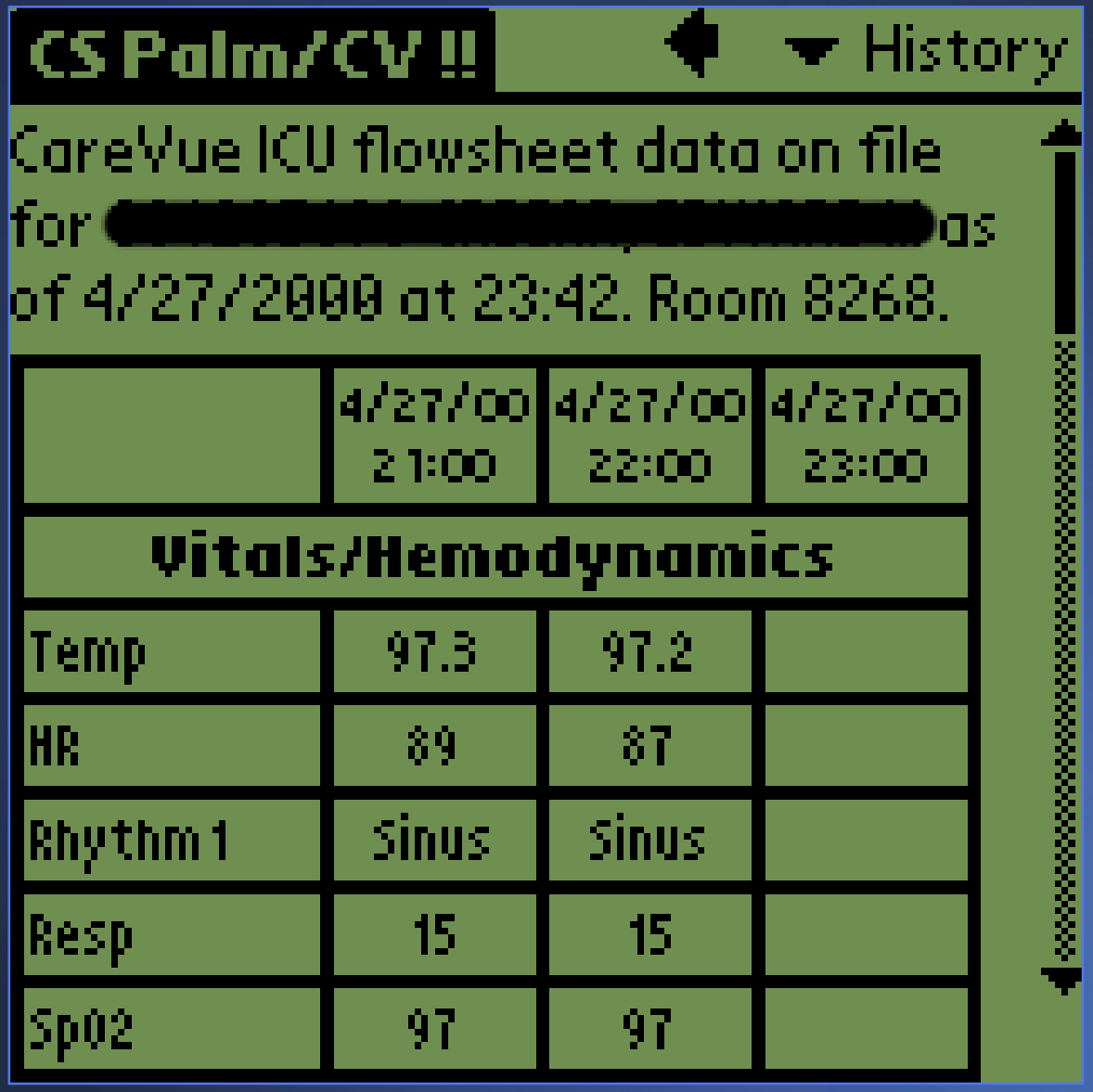
Palm/VS—Transcribed Report List



Palm/VS—Transcribed Report Detail

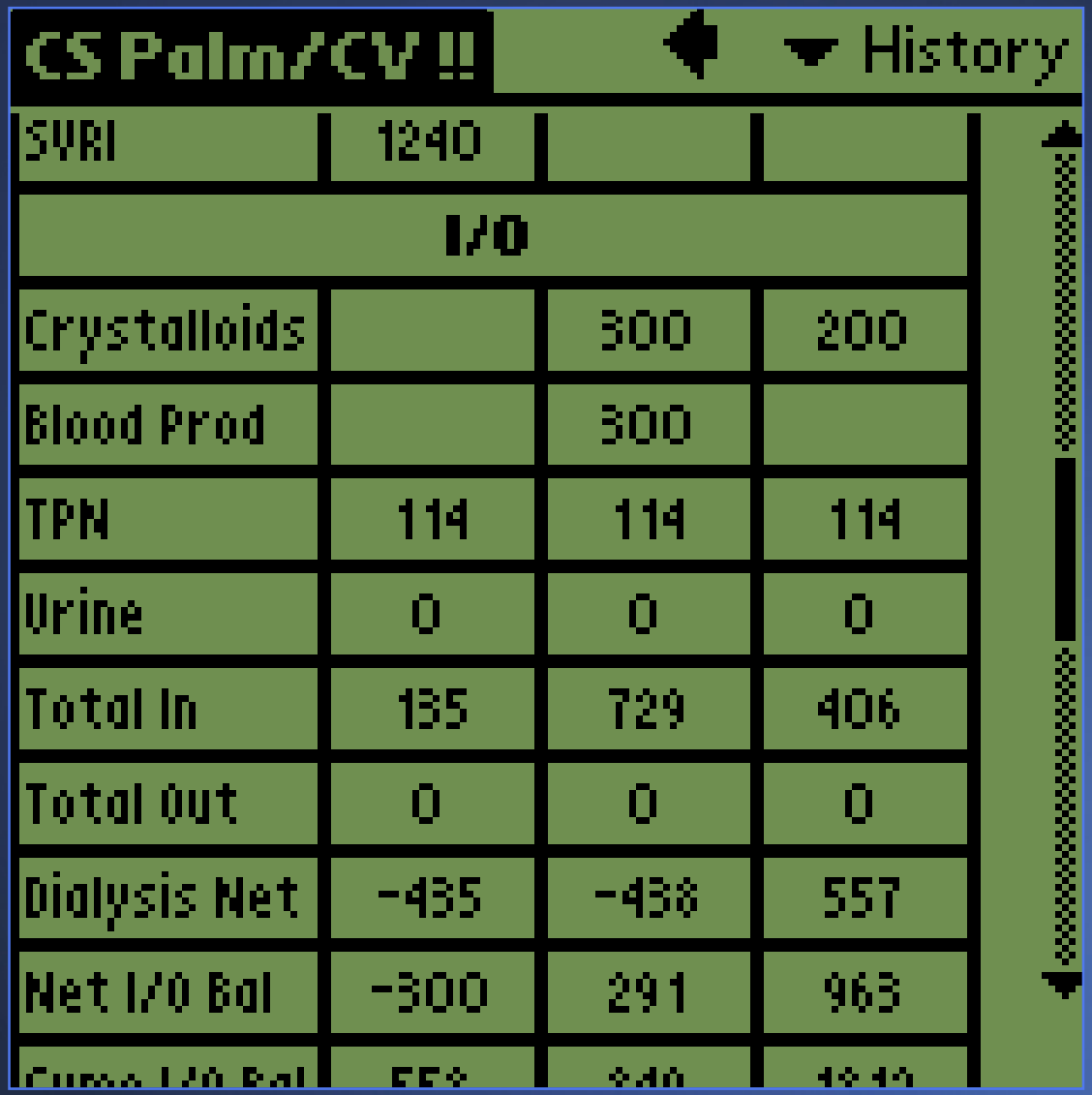


Palm/VS–ICU Vital Signs

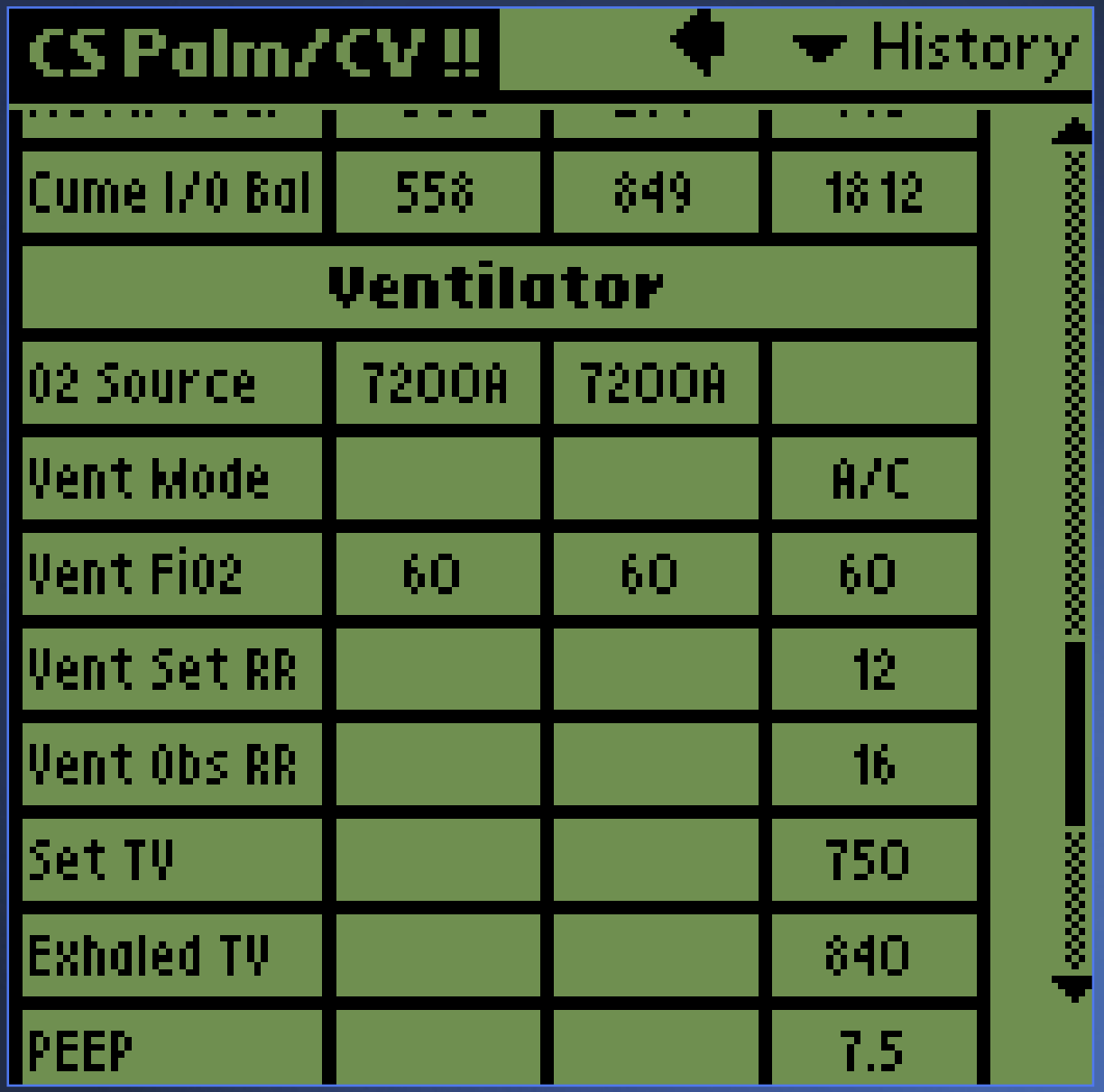


	4/27/00 21:00	4/27/00 22:00	4/27/00 23:00
Vitals/Hemodynamics			
Temp	97.3	97.2	
HR	89	87	
Rhythm 1	Sinus	Sinus	
Resp	15	15	
SpO2	97	97	

Palm/VS—ICU Intake and Output



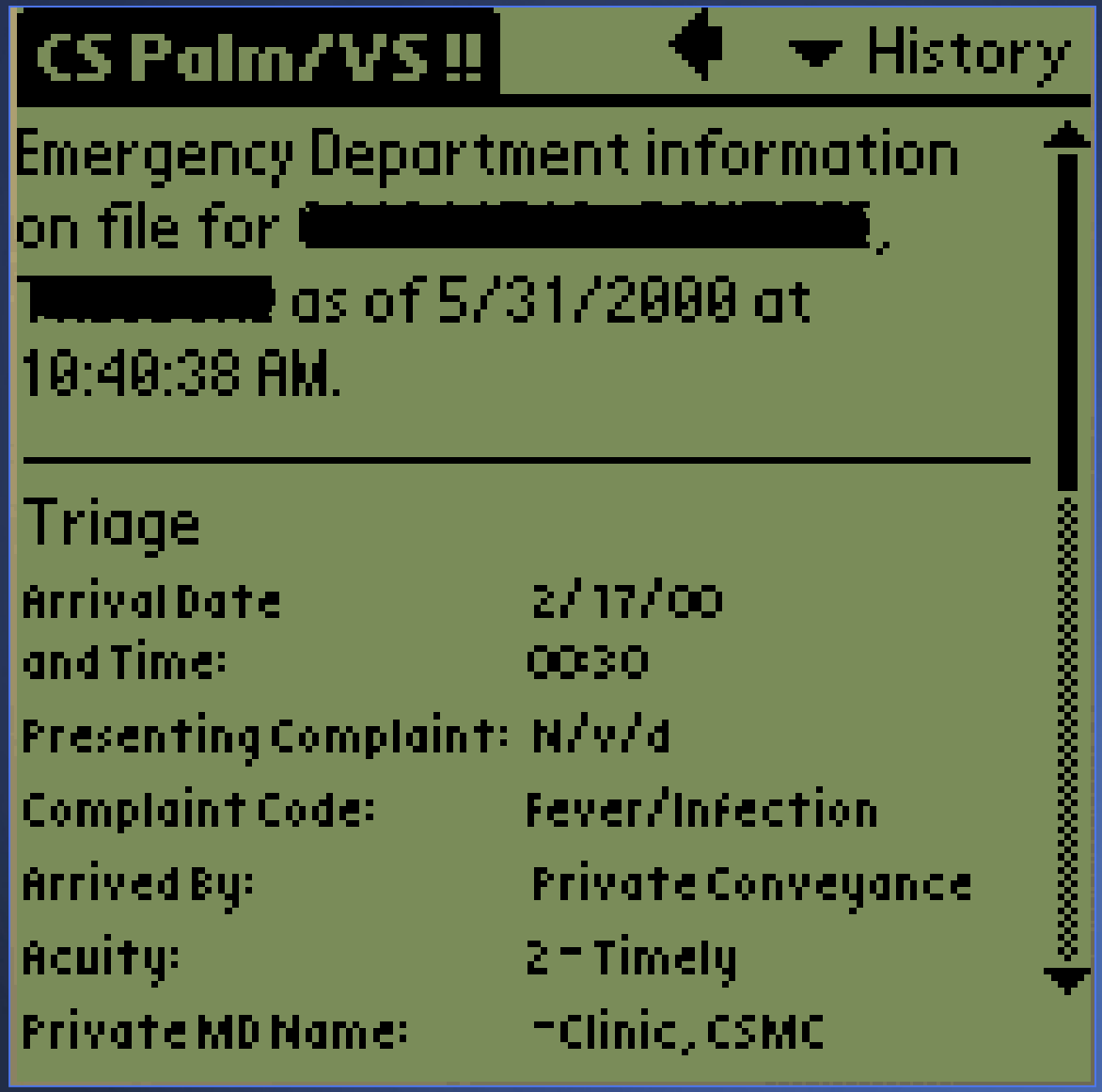
Palm/VS—ICU Ventilator



Palm/VS—ICU Blood Gases



Palm/VS—Emergency Dept.



Other Wireless Applications Provided to CSMC Physician Users

- Developed in-house:
  - Attending Staff Directory
  - Employee Directory
  - Library Document/Book Orders
  - Alphanumeric Paging
- Third-party:
  - MedLine
  - Go Translator
  - POP3 Email Client

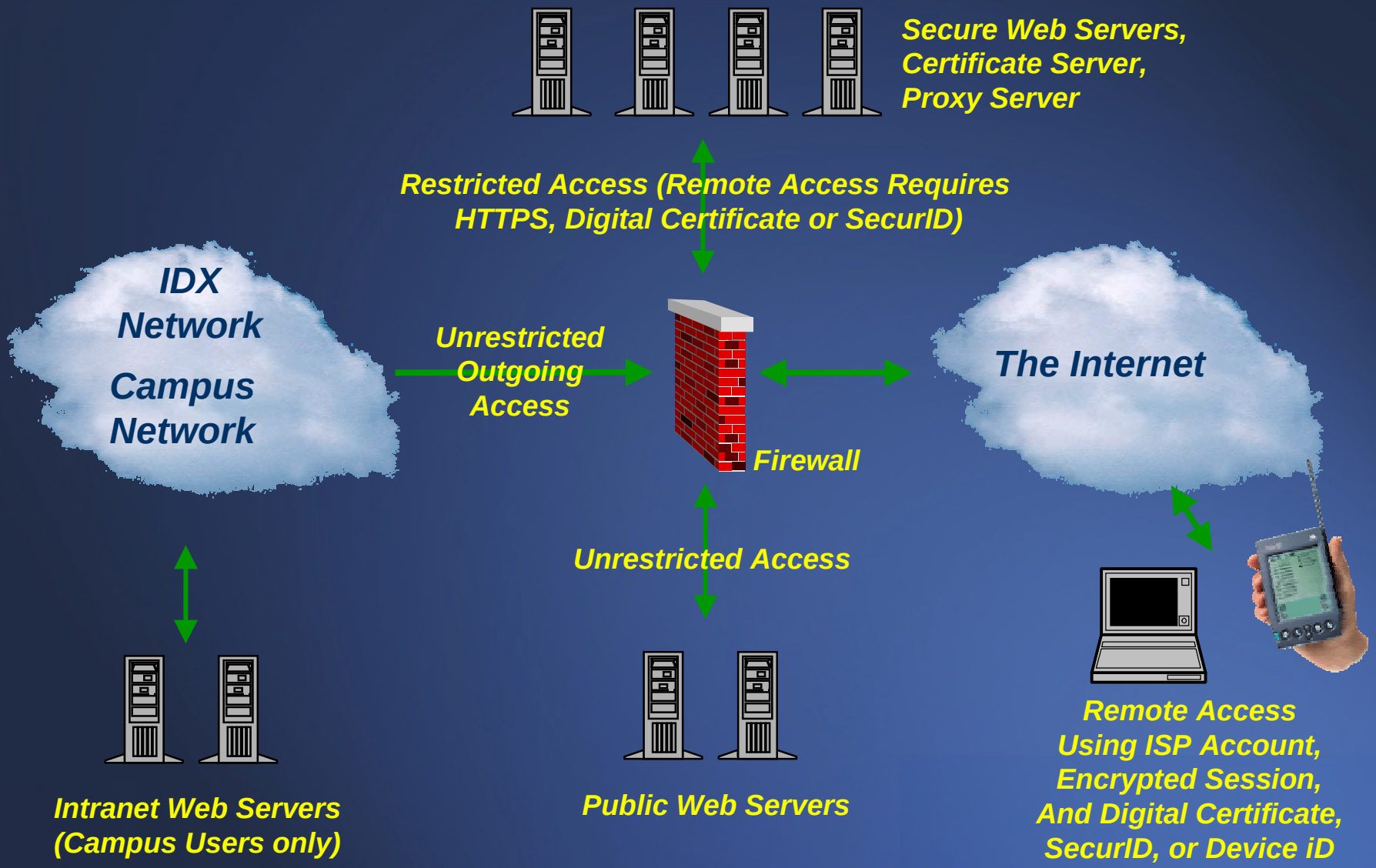
Security and Confidentiality

- Ease-of-Access Stance—Clinicians Can Access (nearly) All Data
- All Employees Sign the HR “Need to Know” Policy
- Close Coordination with Human Resources and Medical Records
- System Design Driven by CSMC Clinical Systems Security Architecture
- Layered Approach—“Defense in Depth”

Defense in Depth

- Encryption
- Firewall and Server Configuration
- Authentication
- Exhaustive Logging of Accesses
- Surveillance Reports
- Ad-Hoc Audit Tools

Intranet and Internet

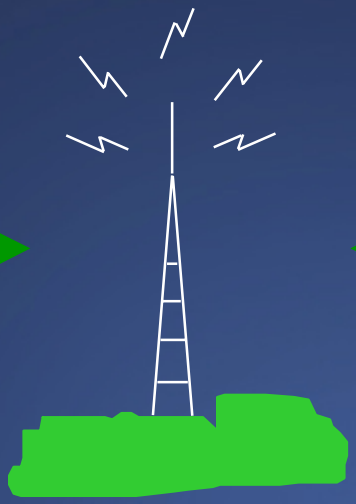


Palm VII Networking Details



DESX

Certicom
Ellyptic Key



DESX

Certicom
Ellyptic Key



Palm.Net Proxy Server

Palm VII

PalmNet (BellSouth) Network

SSL HTTPS



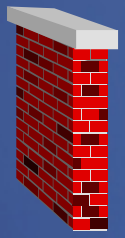
ODBC

TCP/IP



SSL

HTTPS



SSL

HTTPS



CDR Database

CSMC Secure Web Server

CSMC Firewall

The Internet

Encryption

- End to end encryption is used for all wireless transactions
- Certicom elliptic-curve cryptographic technology (DESC) for wireless and private network
- Industry-standard Secure Socket Layer (SSL) for public network

Firewall and Server Security

- Firewall allows only HTTPS (SSL) traffic to Palm VII application server
- Server's virtual directory configured to accept connections only from known wireless Proxy servers
- Application code also checks IP address of incoming connection, rejects unknown IPs

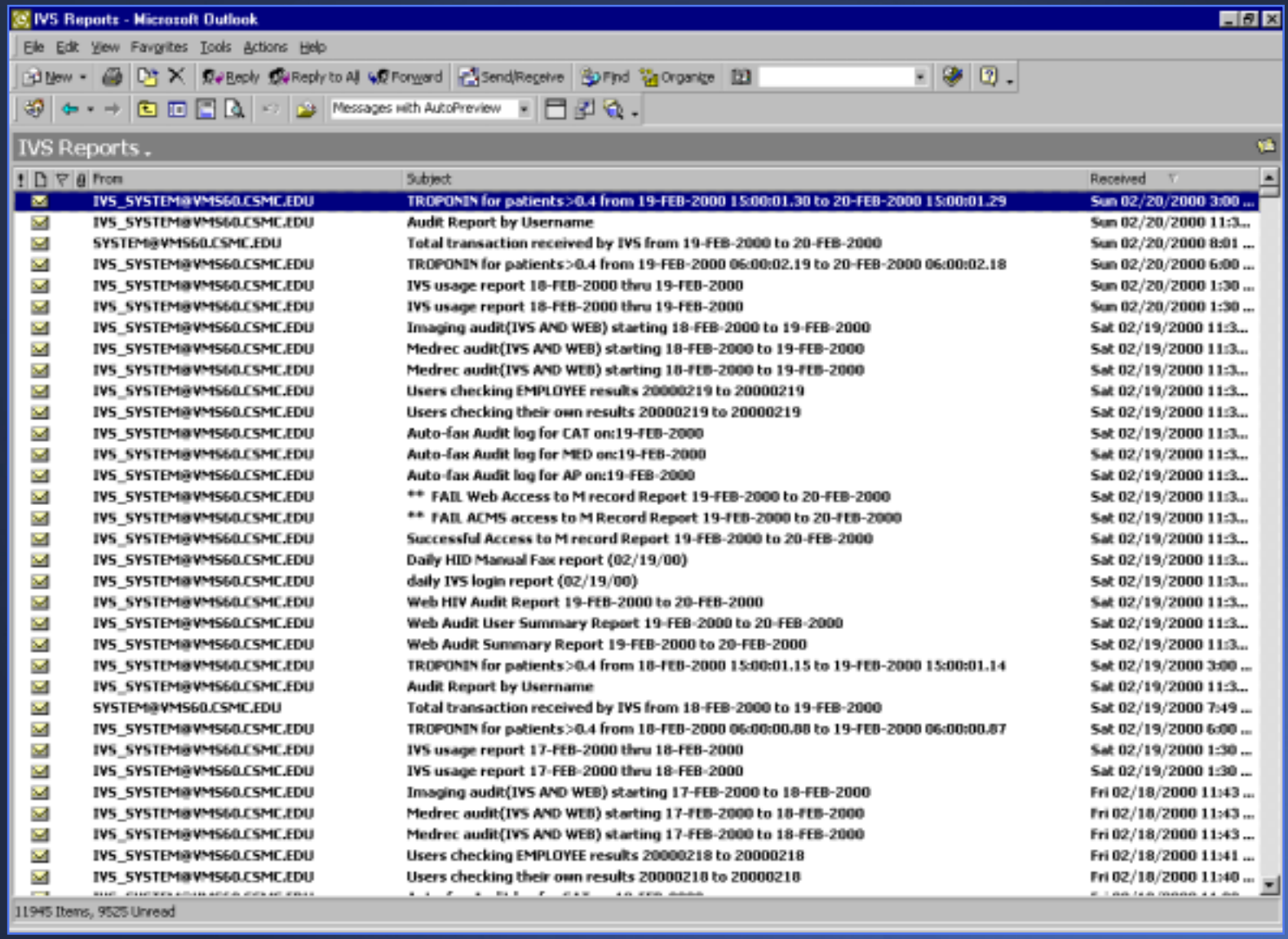
Authentication

- Two-factor authentication—“something you have, and something you know”
- Palm VII unique device ID is passed in all transactions
- Palm application checks and logs device ID, can lock out device if lost or stolen
- User must also enter application user ID and password

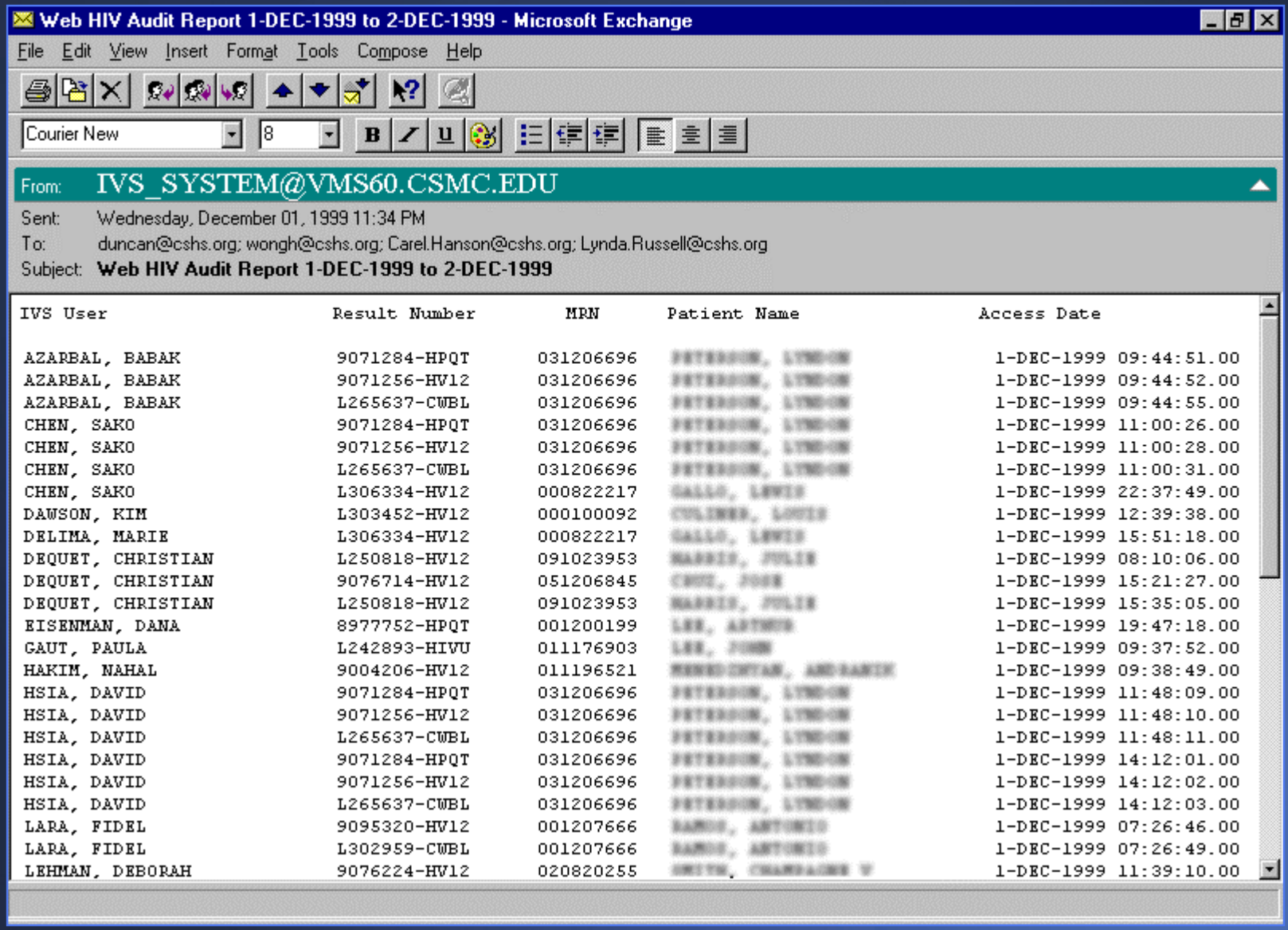
Logging and Surveillance

- Leverages Web/VS security infrastructure
- Every access to clinical data is logged with date and time, user ID, patient ID, data viewed
- Daily surveillance reports for unusual patterns of activity
- Interactive web tool allows ad-hoc audits of patient account or user activity

Daily Surveillance Reports



Sample Surveillance Report (HIV)



Web HIV Audit Report 1-DEC-1999 to 2-DEC-1999 - Microsoft Exchange

File Edit View Insert Format Tools Compose Help

Courier New 8

From: IVS_SYSTEM@VMS60.CSMC.EDU

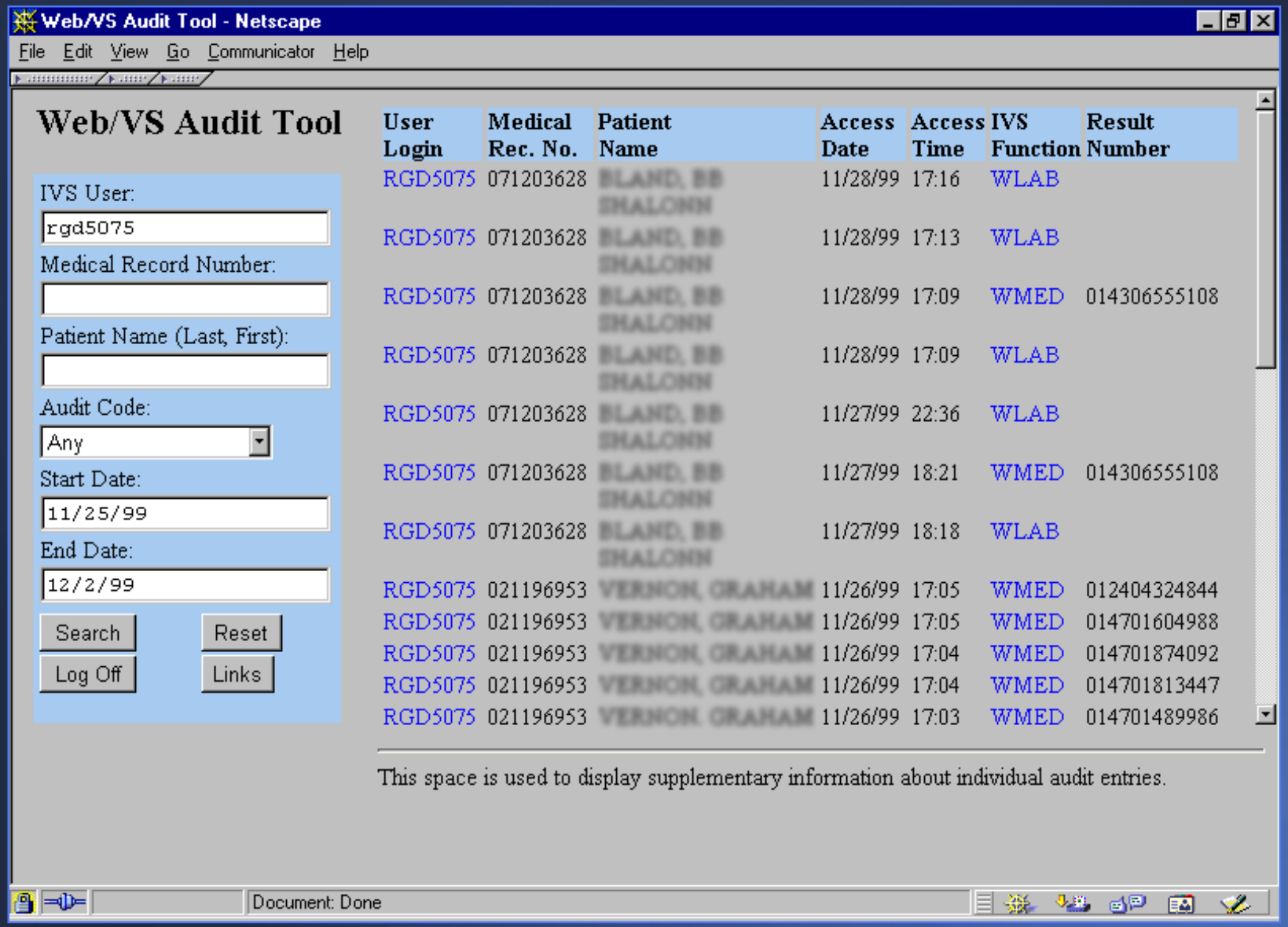
Sent: Wednesday, December 01, 1999 11:34 PM

To: duncan@cshs.org; wongh@cshs.org; Carel.Hanson@cshs.org; Lynda.Russell@cshs.org

Subject: **Web HIV Audit Report 1-DEC-1999 to 2-DEC-1999**

IVS User	Result Number	MRN	Patient Name	Access Date
AZARBAL, BABAK	9071284-HPQT	031206696	[illegible]	1-DEC-1999 09:44:51.00
AZARBAL, BABAK	9071256-HV12	031206696	[illegible]	1-DEC-1999 09:44:52.00
AZARBAL, BABAK	L265637-CWBL	031206696	[illegible]	1-DEC-1999 09:44:55.00
CHEN, SAKO	9071284-HPQT	031206696	[illegible]	1-DEC-1999 11:00:26.00
CHEN, SAKO	9071256-HV12	031206696	[illegible]	1-DEC-1999 11:00:28.00
CHEN, SAKO	L265637-CWBL	031206696	[illegible]	1-DEC-1999 11:00:31.00
CHEN, SAKO	L306334-HV12	000822217	[illegible]	1-DEC-1999 22:37:49.00
DAWSON, KIM	L303452-HV12	000100092	[illegible]	1-DEC-1999 12:39:38.00
DELIMA, MARIE	L306334-HV12	000822217	[illegible]	1-DEC-1999 15:51:18.00
DEQUET, CHRISTIAN	L250818-HV12	091023953	[illegible]	1-DEC-1999 08:10:06.00
DEQUET, CHRISTIAN	9076714-HV12	051206845	[illegible]	1-DEC-1999 15:21:27.00
DEQUET, CHRISTIAN	L250818-HV12	091023953	[illegible]	1-DEC-1999 15:35:05.00
EISENMAN, DANA	8977752-HPQT	001200199	[illegible]	1-DEC-1999 19:47:18.00
GAUT, PAULA	L242893-HIVU	011176903	[illegible]	1-DEC-1999 09:37:52.00
HAKIM, NAHAL	9004206-HV12	011196521	[illegible]	1-DEC-1999 09:38:49.00
HSIA, DAVID	9071284-HPQT	031206696	[illegible]	1-DEC-1999 11:48:09.00
HSIA, DAVID	9071256-HV12	031206696	[illegible]	1-DEC-1999 11:48:10.00
HSIA, DAVID	L265637-CWBL	031206696	[illegible]	1-DEC-1999 11:48:11.00
HSIA, DAVID	9071284-HPQT	031206696	[illegible]	1-DEC-1999 14:12:01.00
HSIA, DAVID	9071256-HV12	031206696	[illegible]	1-DEC-1999 14:12:02.00
HSIA, DAVID	L265637-CWBL	031206696	[illegible]	1-DEC-1999 14:12:03.00
LARA, FIDEL	9095320-HV12	001207666	[illegible]	1-DEC-1999 07:26:46.00
LARA, FIDEL	L302959-CWBL	001207666	[illegible]	1-DEC-1999 07:26:49.00
LEHMAN, DEBORAH	9076224-HV12	020820255	[illegible]	1-DEC-1999 11:39:10.00

Interactive Web Audit Tool



Palm VII Issues

- Bandwidth 8 Kbps is comparable to slow modem
- Memory—2 MB in Palm VII, 8 MB in Palm VIIx
- No JavaScript, “cookies”, frames, nested tables
- Screen size (160x160), no horizontal scrolling
- Limited fonts, limited graphics, only 4 “colors”
- Signal is poor in some buildings and underground
- Handwritten entry of passwords
- Page cache persists across sessions
- Ongoing Palm.Net® wireless costs and account management


accelerate
collaborate
innovate
communicate
palm
PalmSource
Conference 2000